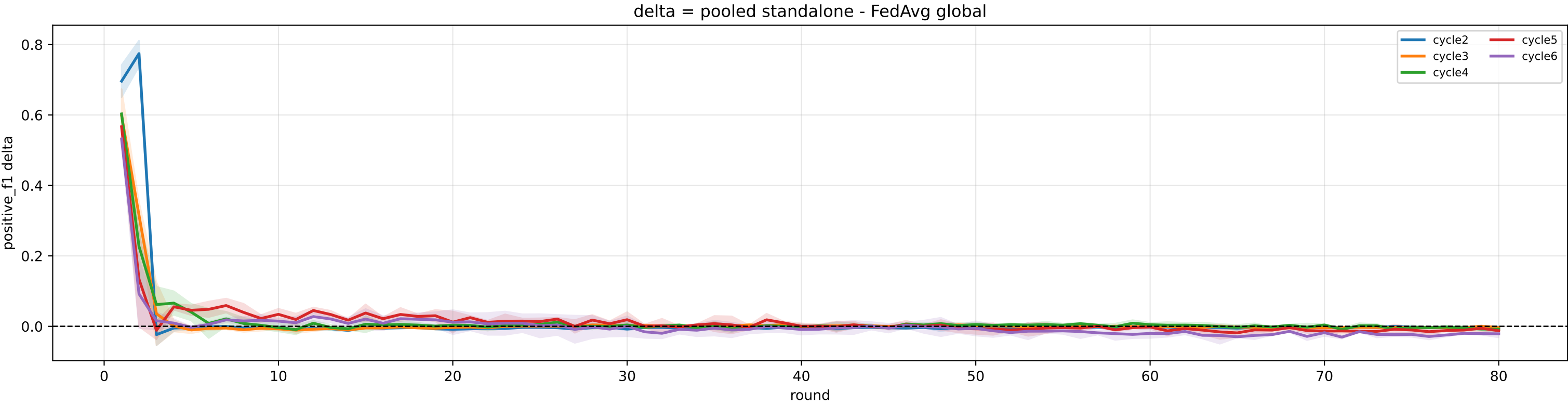
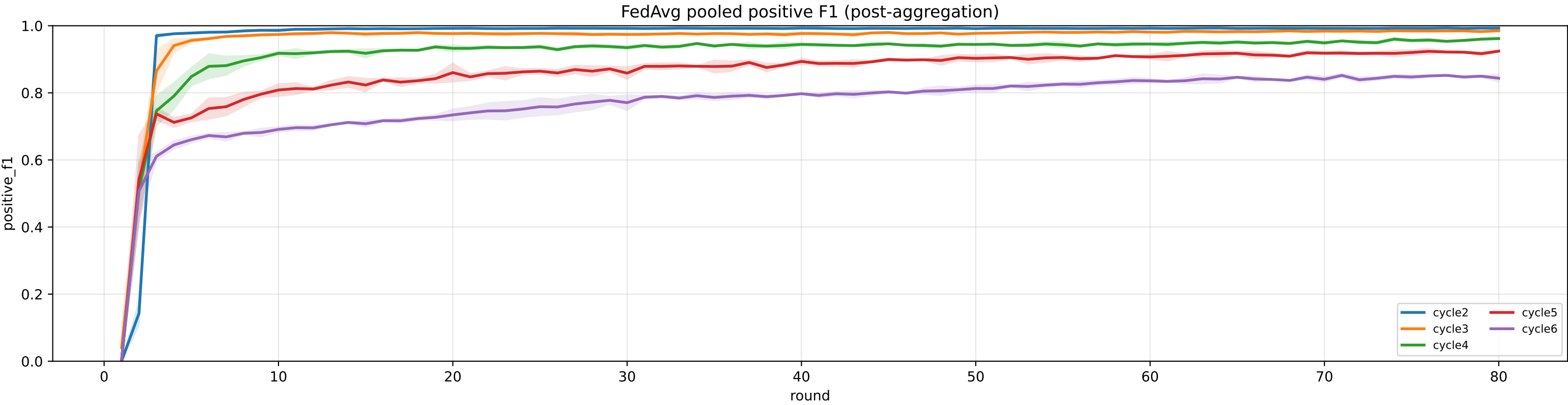
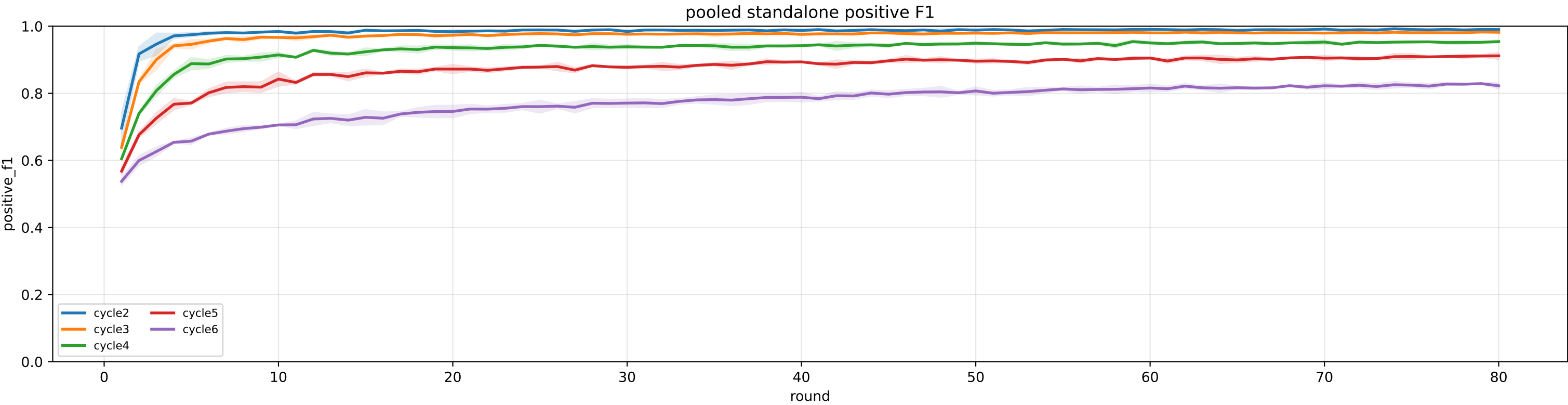
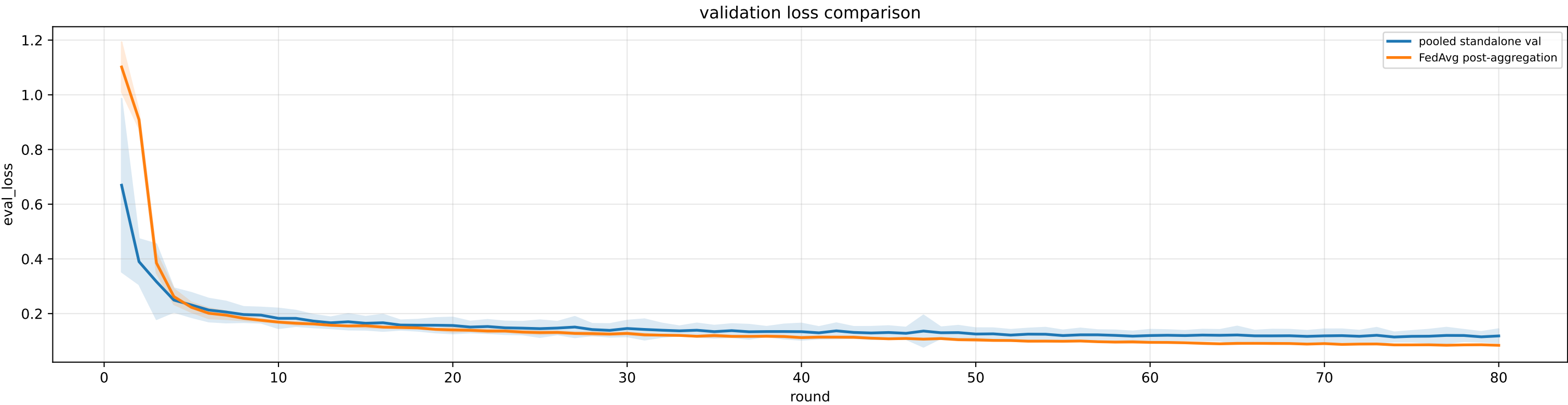


family=strong_decreasing | cycle2=37.8%±0.4% | cycle3=32.8%±0.8% | cycle4=27.9%±0.5% | cycle5=22.4%±0.5% | cycle6=16.0%±0.2%
family=mild_decreasing | cycle2=17.1%±0.7% | cycle3=14.6%±1.1% | cycle4=12.6%±0.5% | cycle5=9.5%±0.8% | cycle6=7.7%±0.4%
family=flat_balanced | cycle2=8.6%±0.7% | cycle3=10.2%±0.8% | cycle4=9.7%±1.5% | cycle5=9.3%±1.9% | cycle6=8.6%±0.7%
family=mild_increasing | cycle2=20.6%±0.6% | cycle3=24.0%±0.4% | cycle4=26.6%±0.7% | cycle5=29.1%±0.4% | cycle6=28.9%±0.9%
family=strong_increasing | cycle2=13.6%±0.4% | cycle3=18.0%±1.5% | cycle4=21.8%±0.4% | cycle5=26.1%±0.7% | cycle6=29.3%±0.4%

Best test summary (FedAvg vs Local)

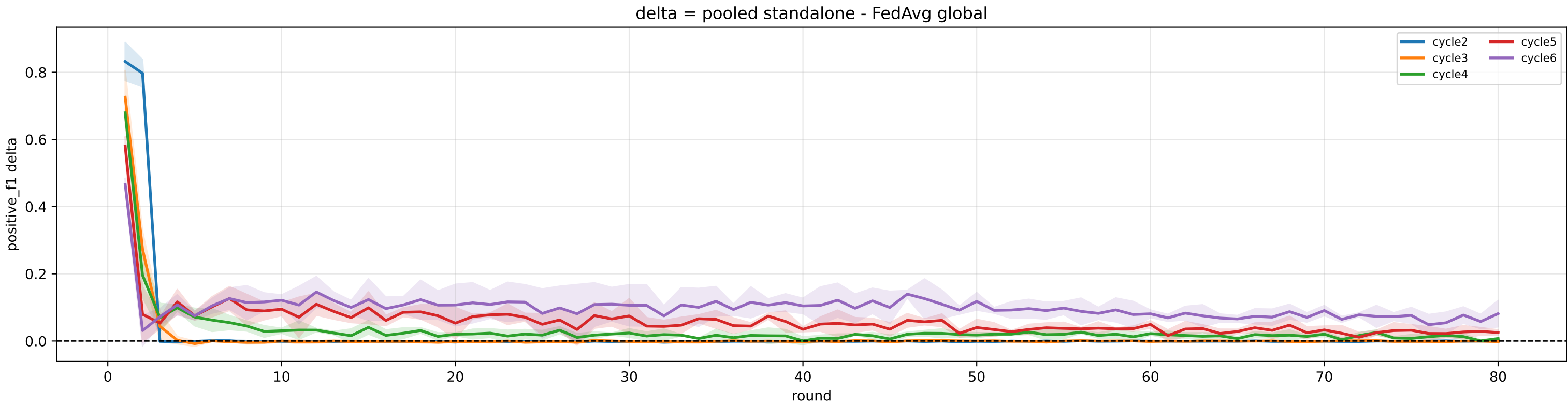
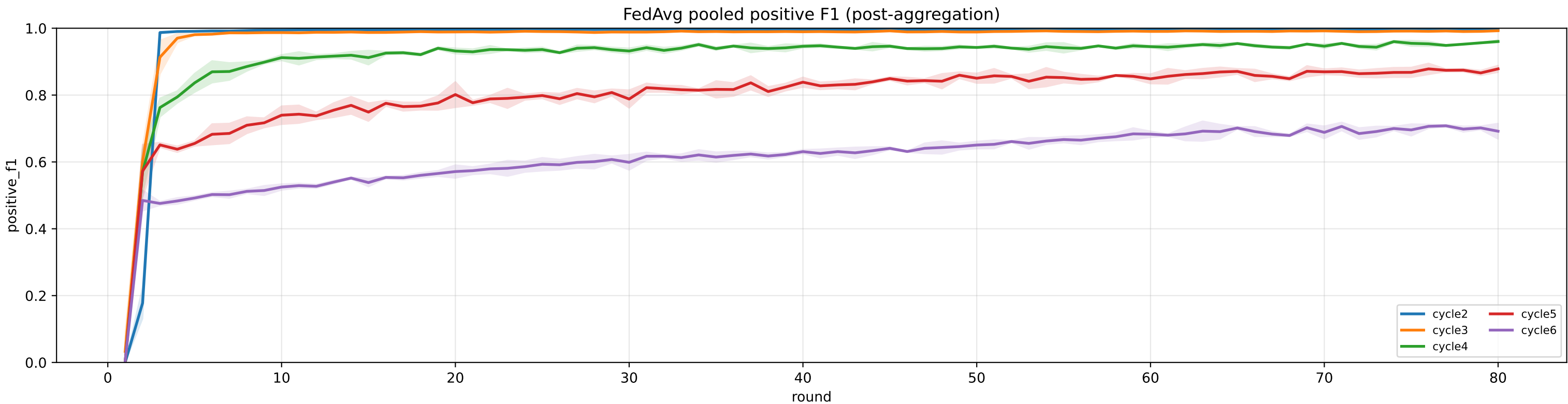
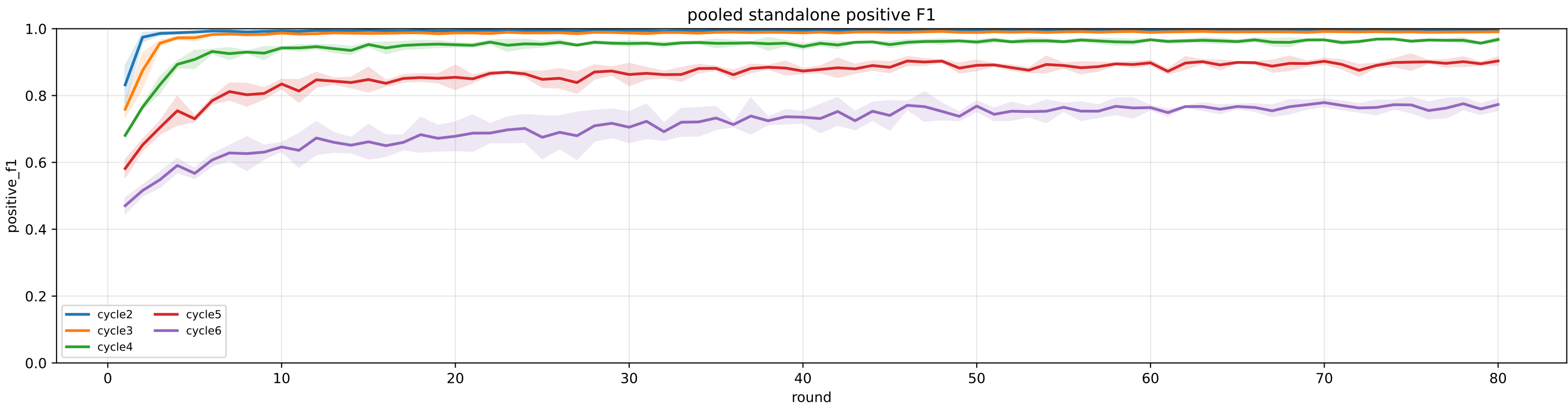
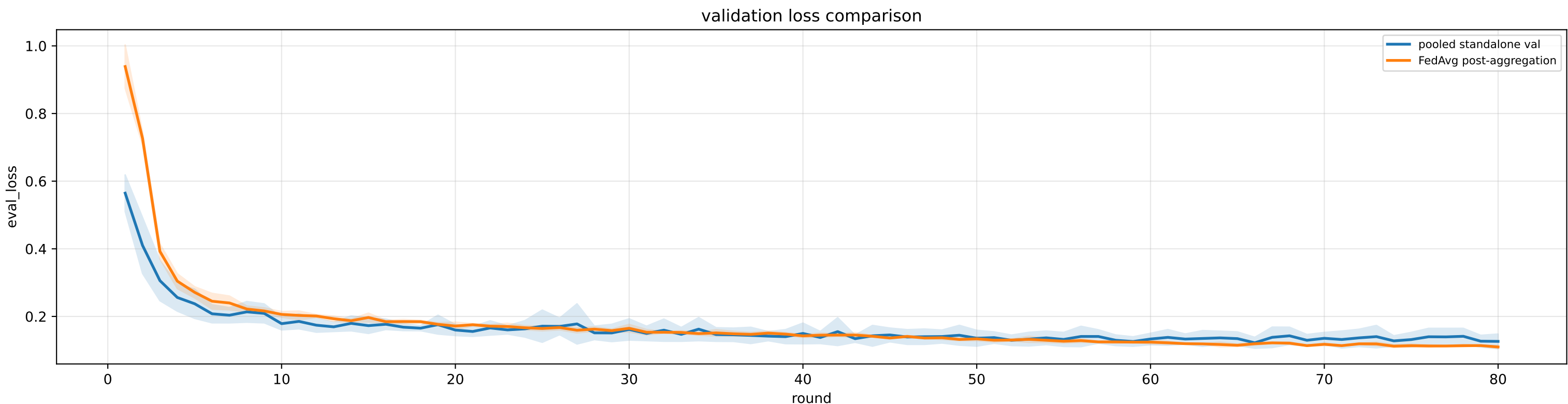
Algorithm	Subset Acc	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	88.9% ± 0.3%	92.3% ± 0.2%	92.1% ± 0.2%	-	-	-	-	-
FedAvg	90.3% ± 0.3%	94.2% ± 0.2%	93.8% ± 0.2%	-	-	-	-	-



family=strong_decreasing | cycle2=37.8%±0.4% | cycle3=32.8%±0.8% | cycle4=27.9%±0.5% | cycle5=22.4%±0.5% | cycle6=16.0%±0.2%

Best test summary (FedAvg vs Local)

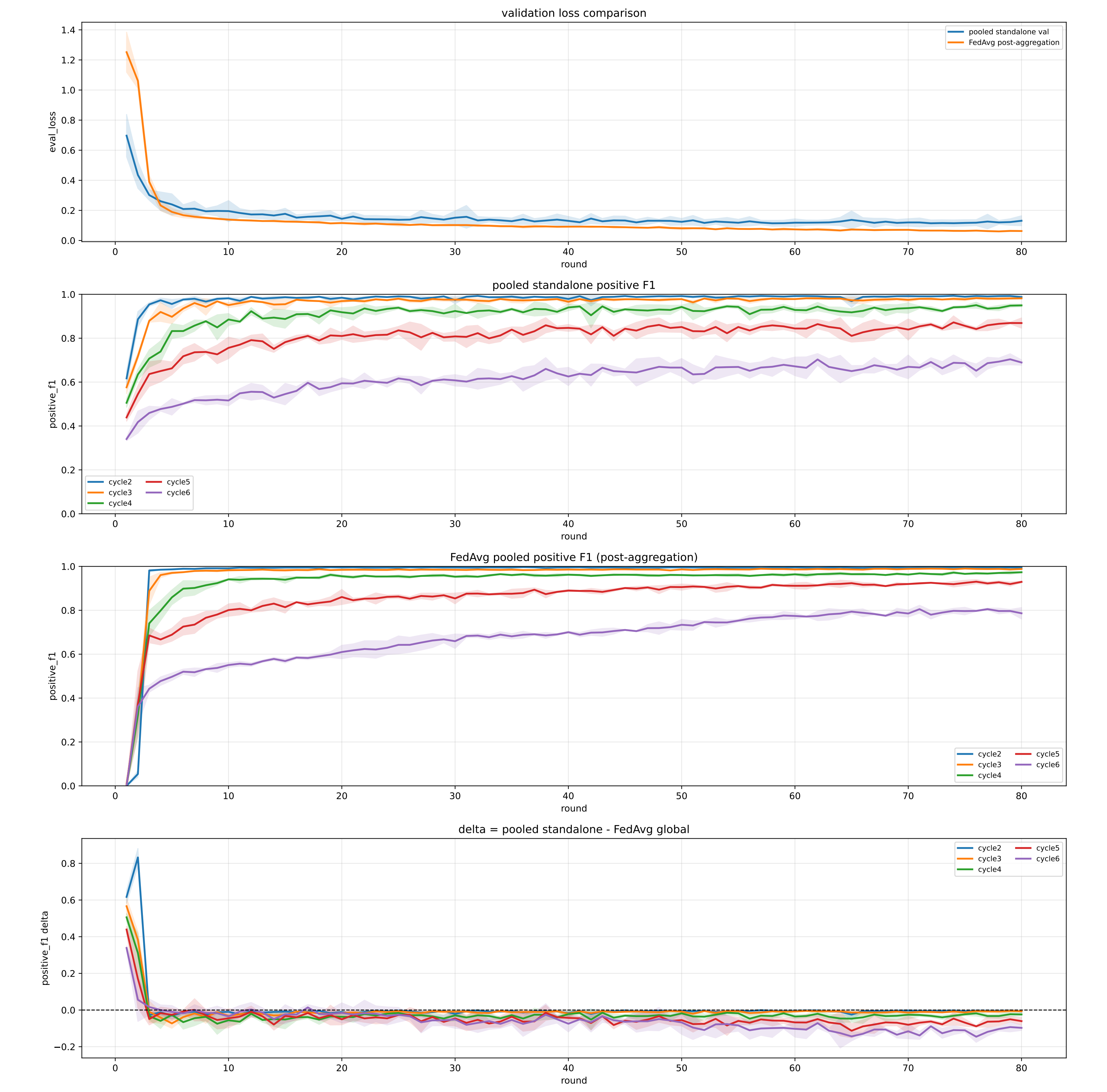
Algorithm	Subset Acc	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	87.5% ± 0.4%	94.5% ± 0.2%	92.5% ± 0.3%	-	-	-	-	-
FedAvg	83.8% ± 0.7%	92.7% ± 0.3%	90.6% ± 0.3%	-	-	-	-	-



family=mild_decreasing | cycle2=17.1%±0.7% | cycle3=14.6%±1.1% | cycle4=12.6%±0.5% | cycle5=9.5%±0.8% | cycle6=7.7%±0.4%

Best test summary (FedAvg vs Local)

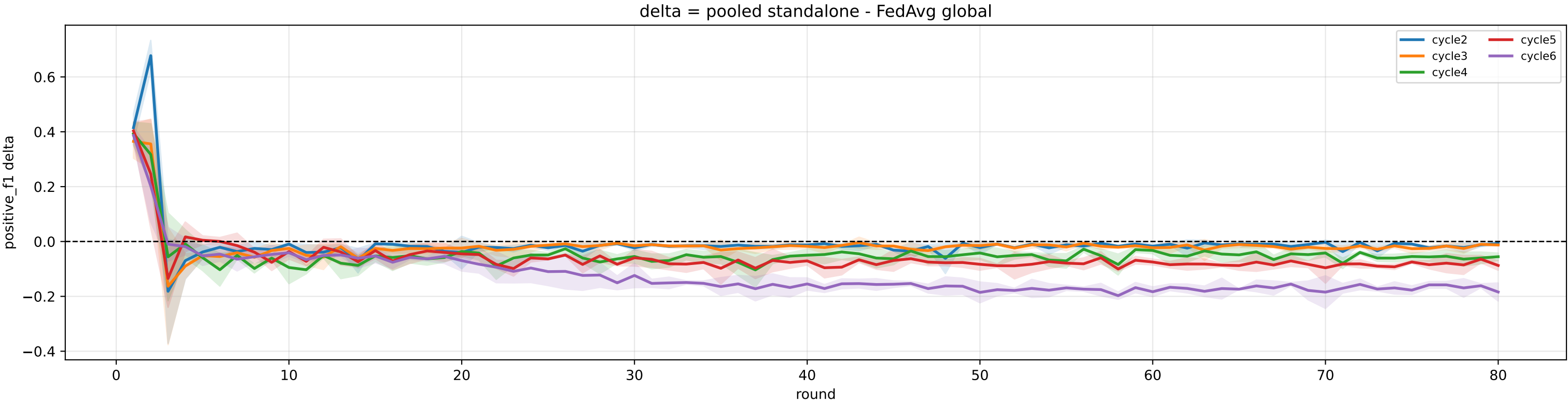
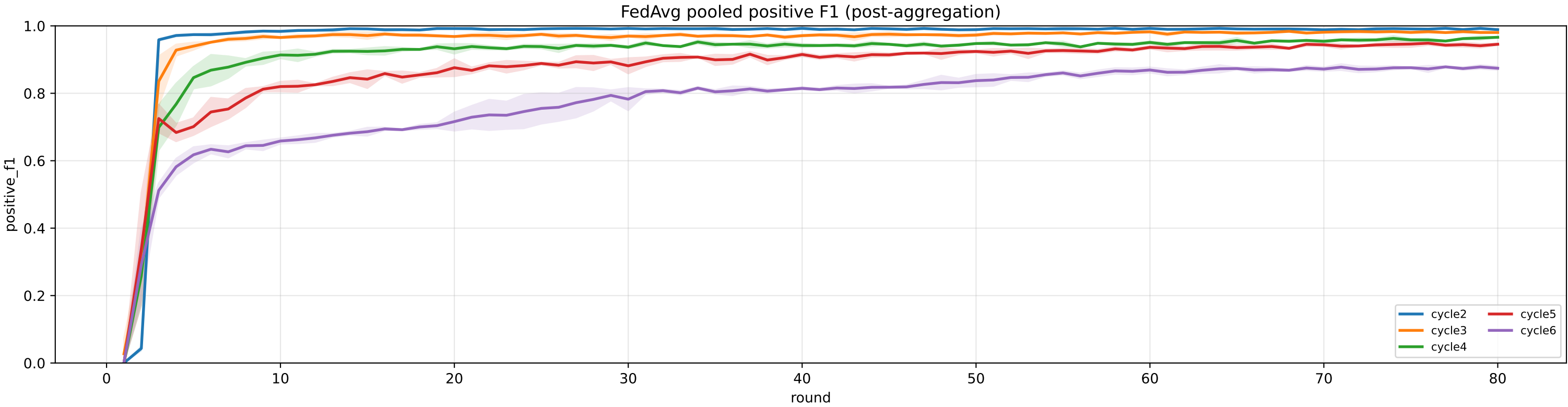
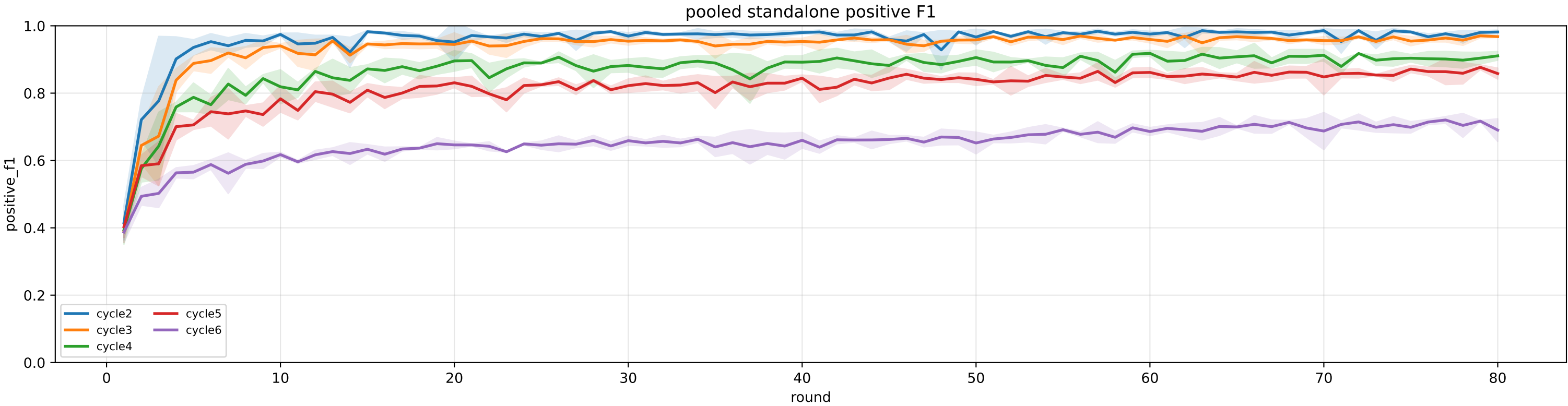
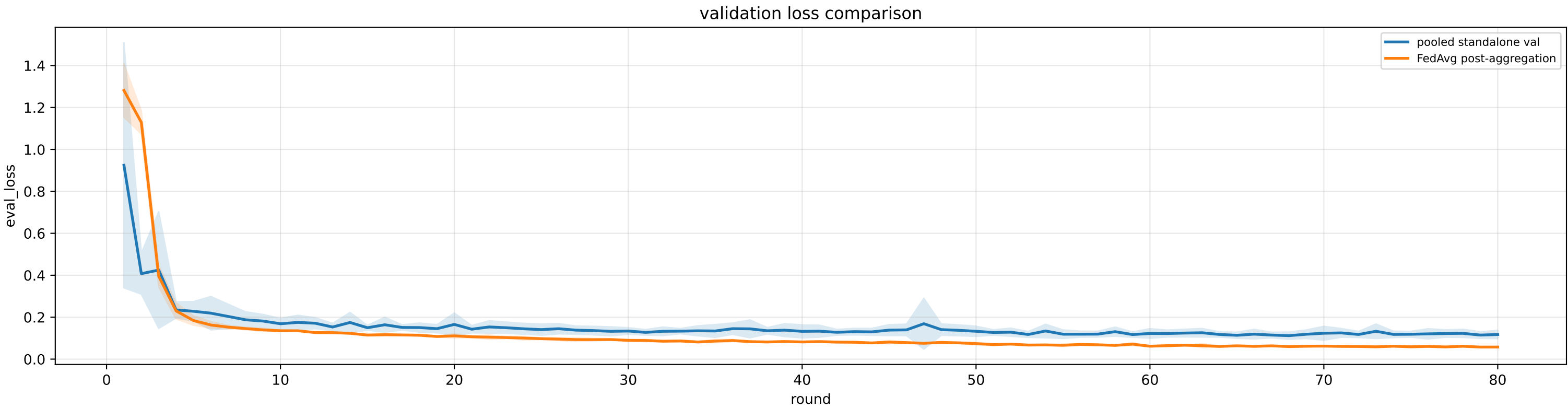
Algorithm	Subset Acc	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	90.6% ± 1.3%	91.1% ± 1.1%	89.3% ± 1.1%	-	-	-	-	-
FedAvg	94.8% ± 0.3%	95.0% ± 0.3%	93.5% ± 0.3%	-	-	-	-	-



family=flat_balanced | cycle2=8.6%±0.7% | cycle3=10.2%±0.8% | cycle4=9.7%±1.5% | cycle5=9.3%±1.9% | cycle6=8.6%±0.7%

Best test summary (FedAvg vs Local)

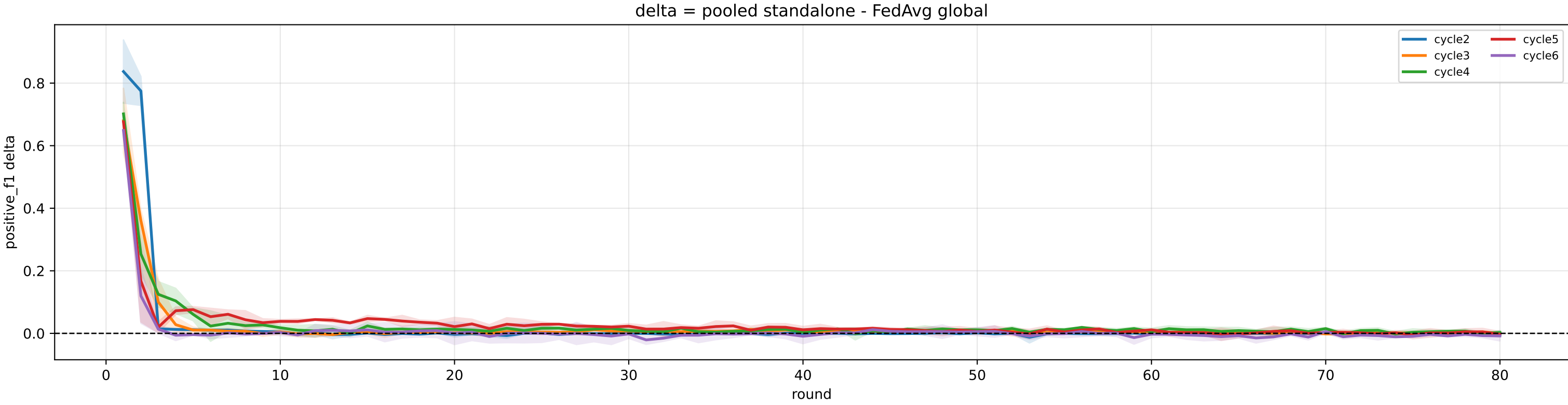
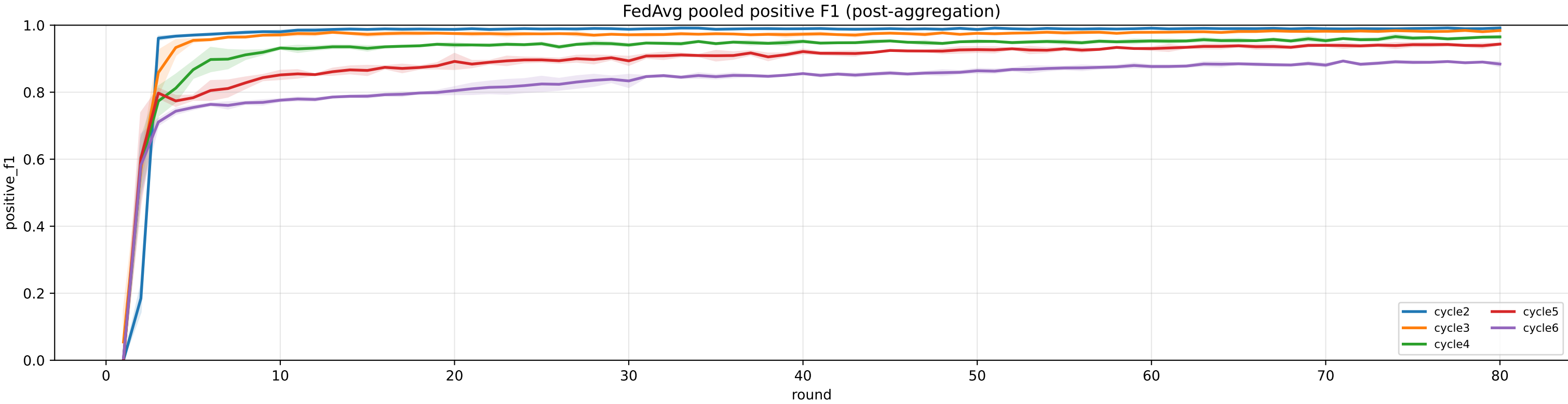
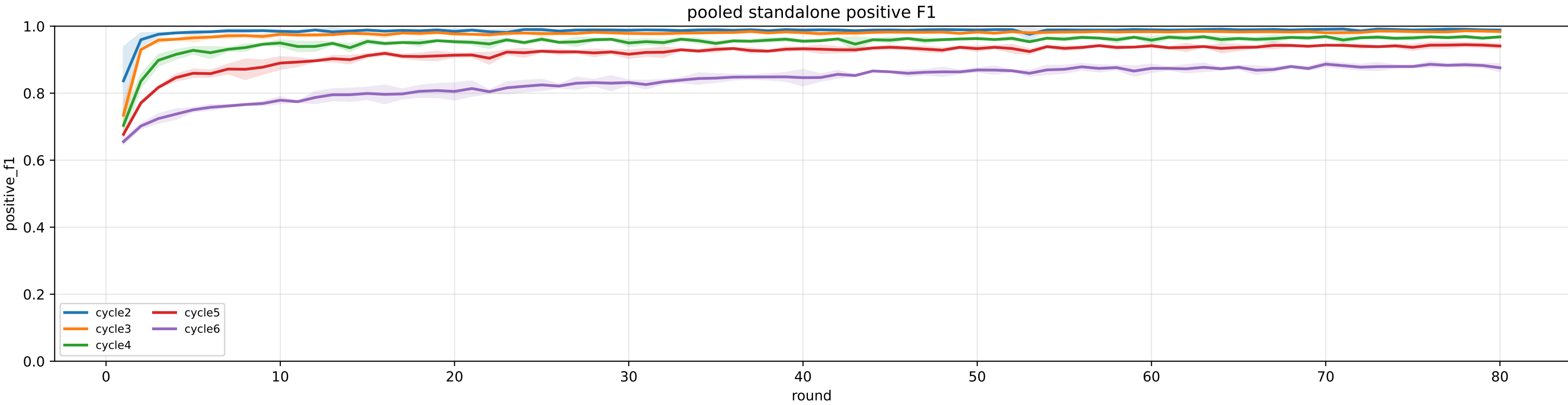
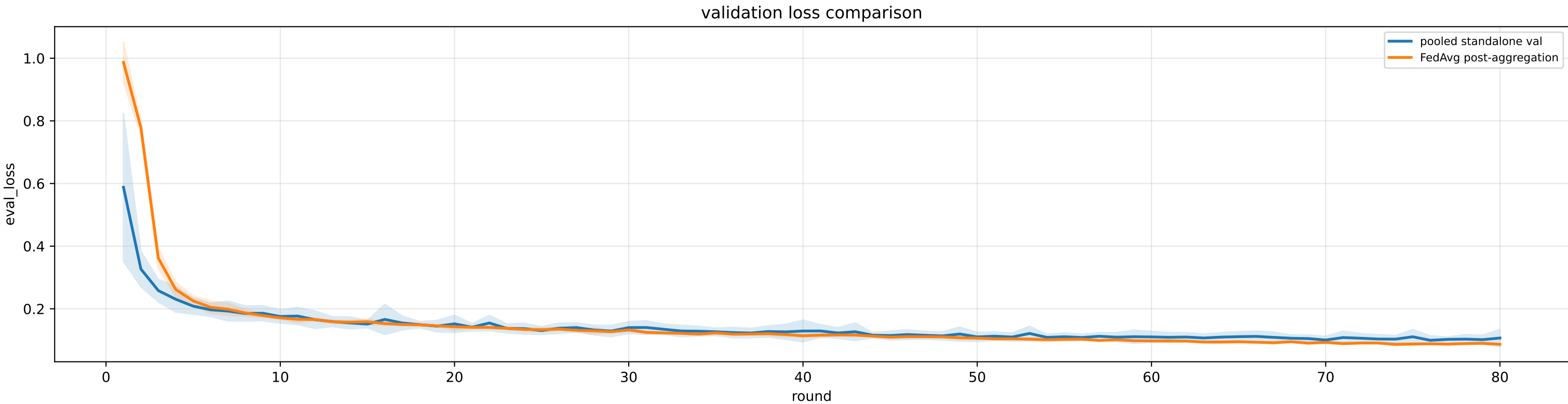
Algorithm	Subset Acc	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	89.7% ± 0.2%	87.6% ± 0.2%	88.9% ± 0.1%	-	-	-	-	-
FedAvg	95.8% ± 0.2%	94.9% ± 0.2%	95.0% ± 0.2%	-	-	-	-	-



family=mild_increasing | cycle2=20.6%±0.6% | cycle3=24.0%±0.4% | cycle4=26.6%±0.7% | cycle5=29.1%±0.4% | cycle6=28.9%±0.9%

Best test summary (FedAvg vs Local)

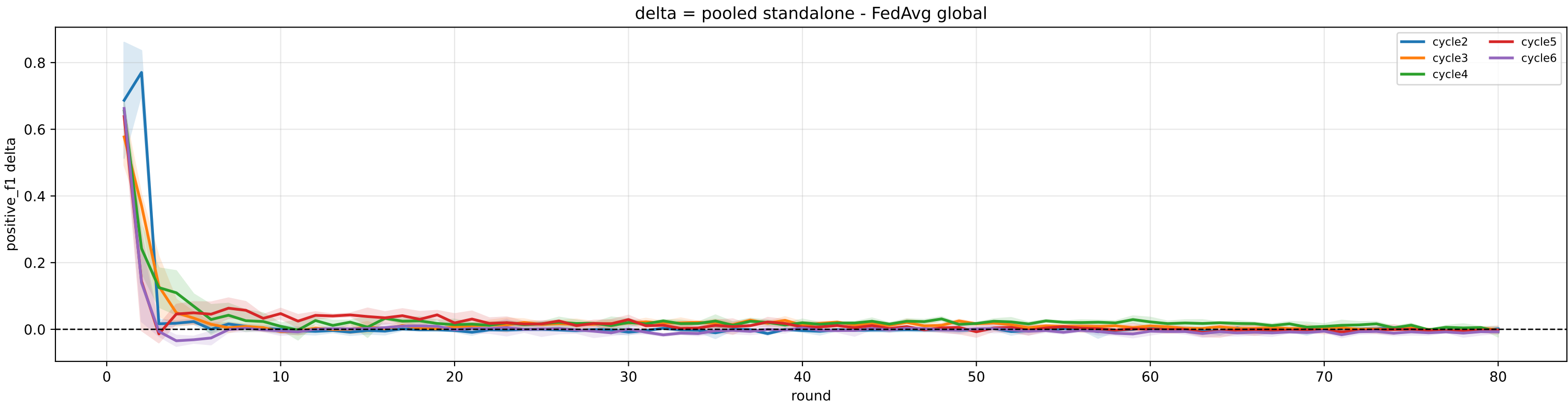
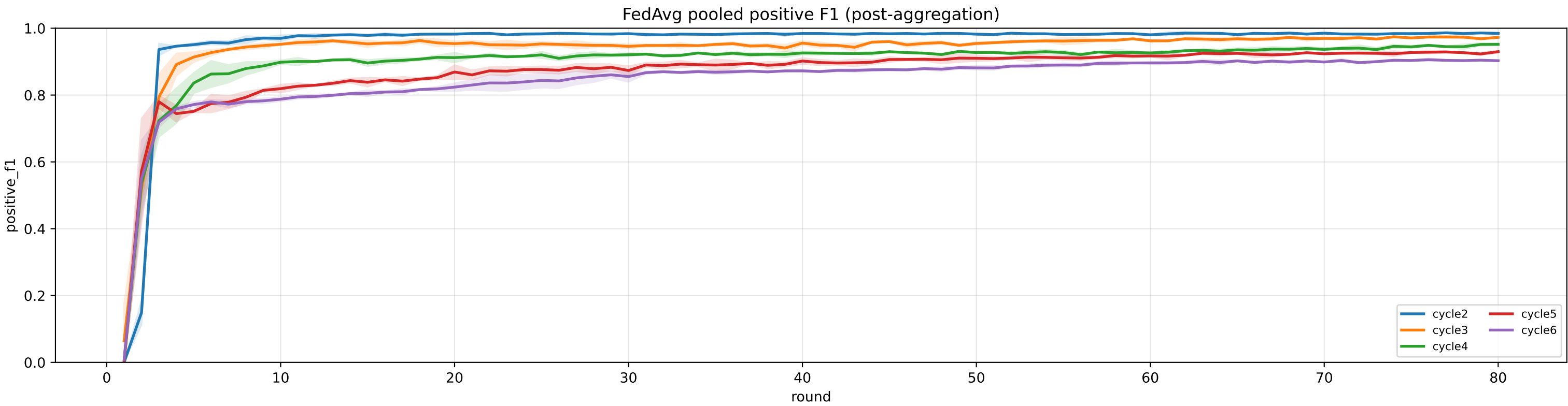
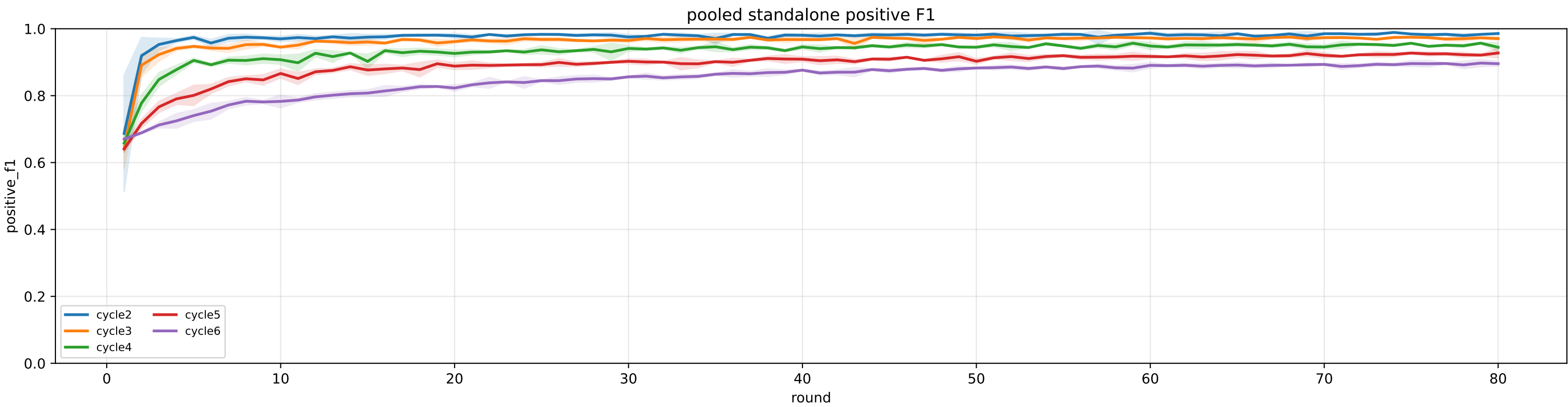
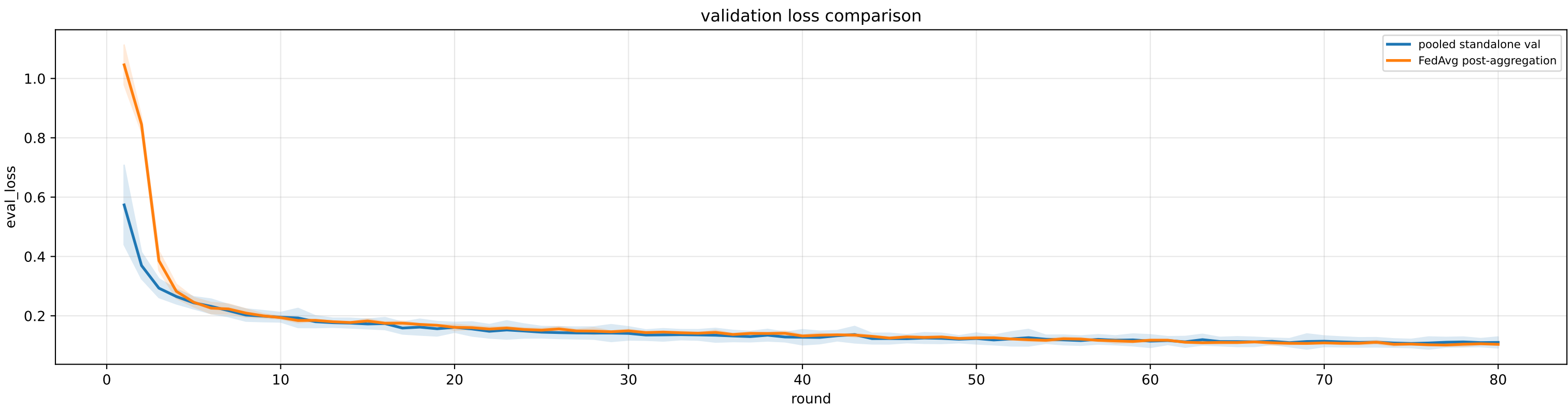
Algorithm	Subset Acc	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	88.9% ± 0.7%	94.9% ± 0.2%	95.3% ± 0.2%	-	-	-	-	-
FedAvg	88.8% ± 0.5%	94.9% ± 0.2%	95.3% ± 0.2%	-	-	-	-	-



family=strong_increasing | cycle2=13.6%±0.4% | cycle3=18.0%±1.5% | cycle4=21.8%±0.4% | cycle5=26.1%±0.7% | cycle6=29.3%±0.4%

Best test summary (FedAvg vs Local)

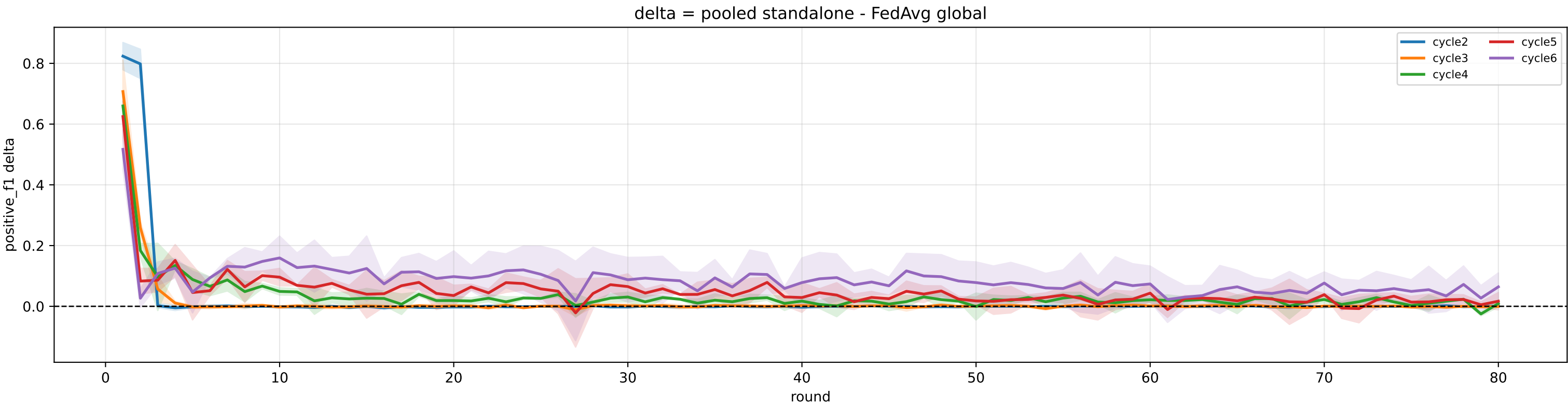
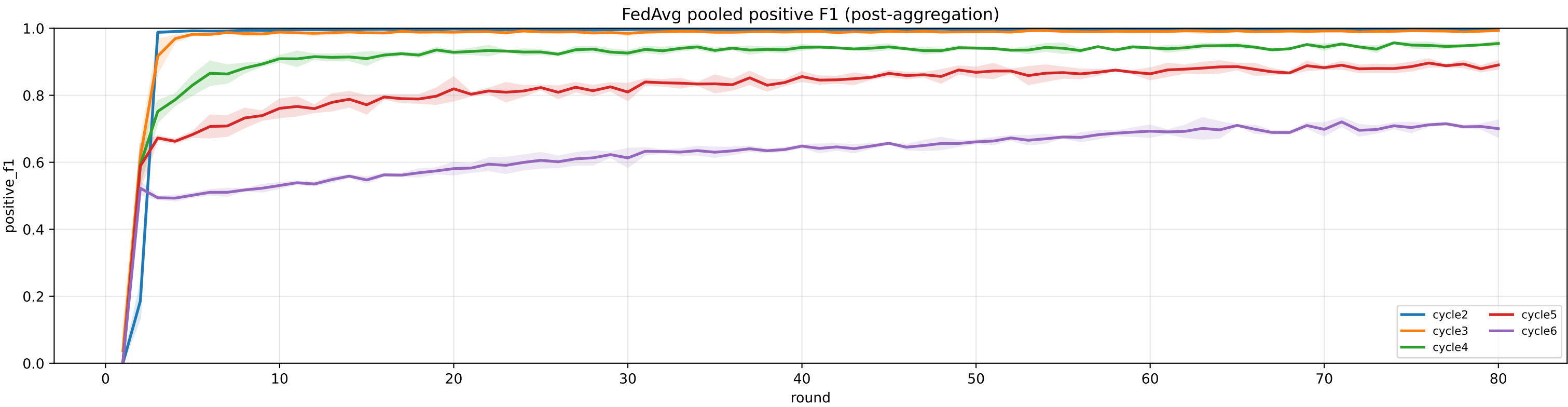
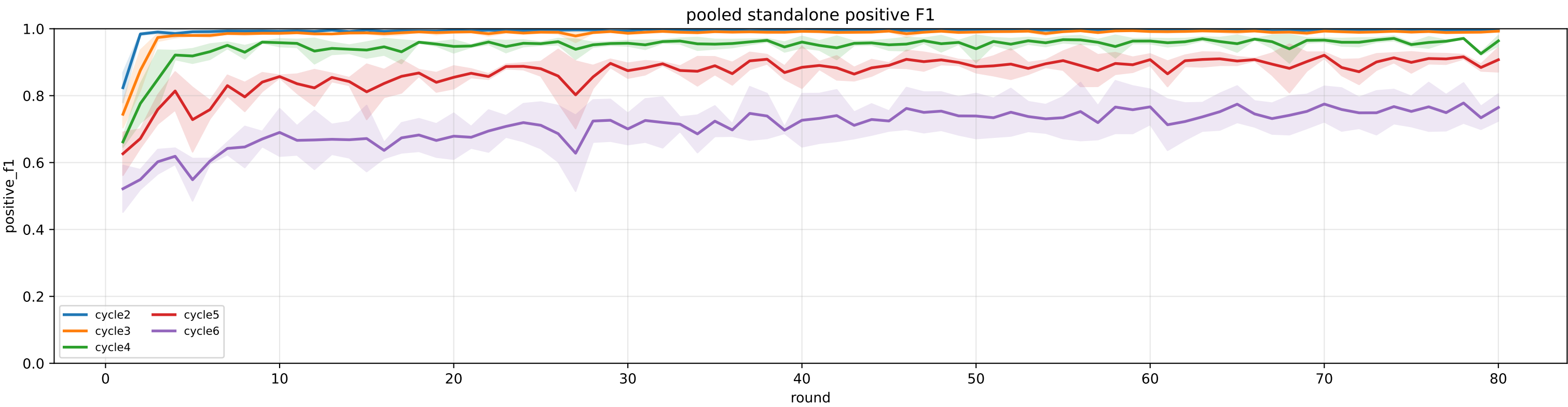
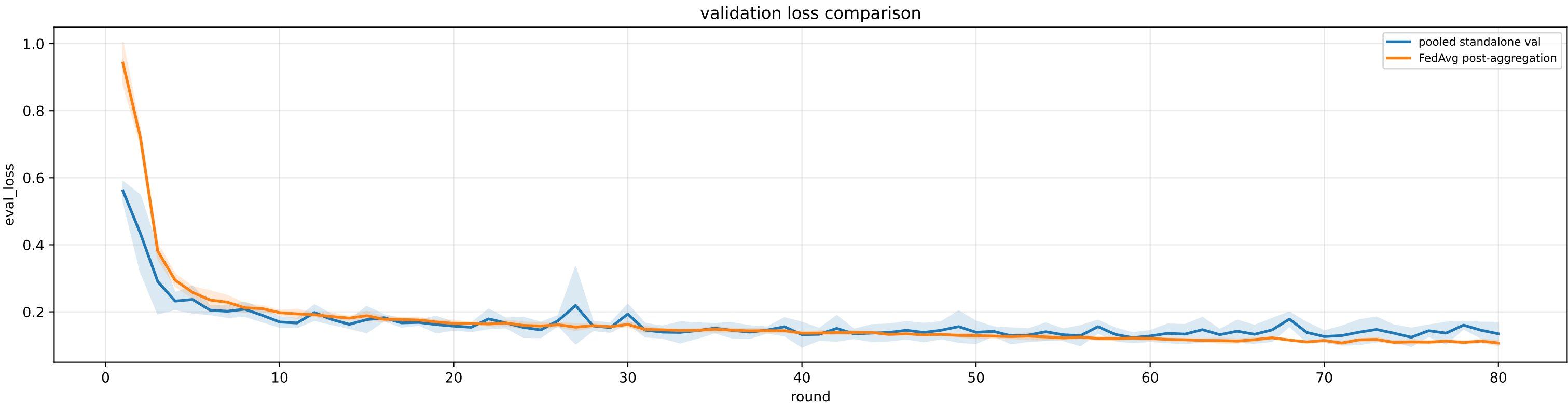
Algorithm	Subset Acc	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	87.7% ± 0.5%	93.6% ± 0.2%	94.6% ± 0.1%	-	-	-	-	-
FedAvg	88.0% ± 0.0%	93.7% ± 0.0%	94.6% ± 0.0%	-	-	-	-	-



graph_id=65 | cycle2=37.3% | cycle3=32.3% | cycle4=28.0% | cycle5=21.8% | cycle6=15.7%

Best test summary (FedAvg vs Local)

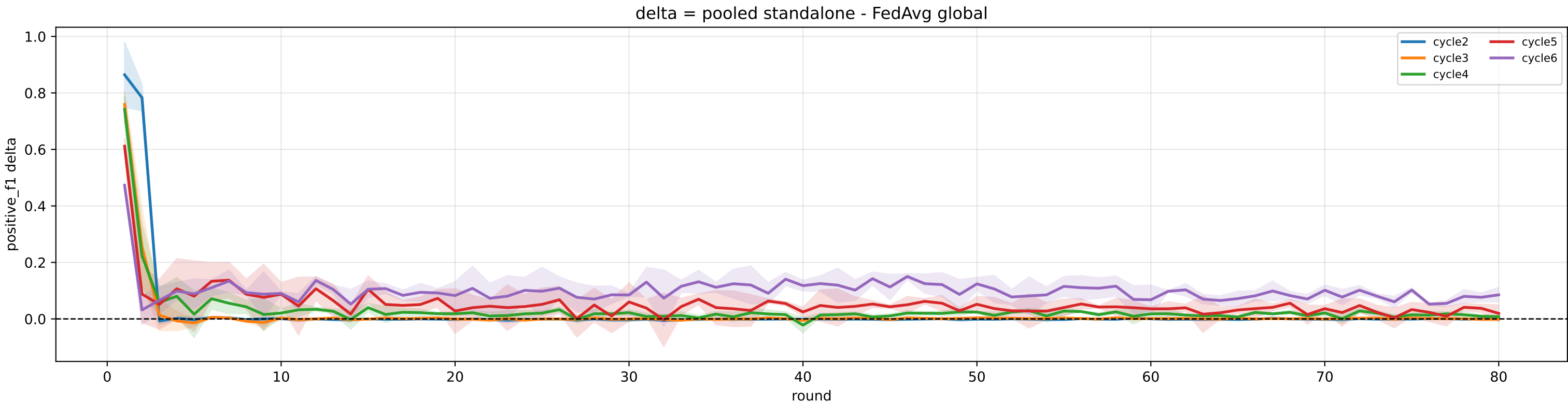
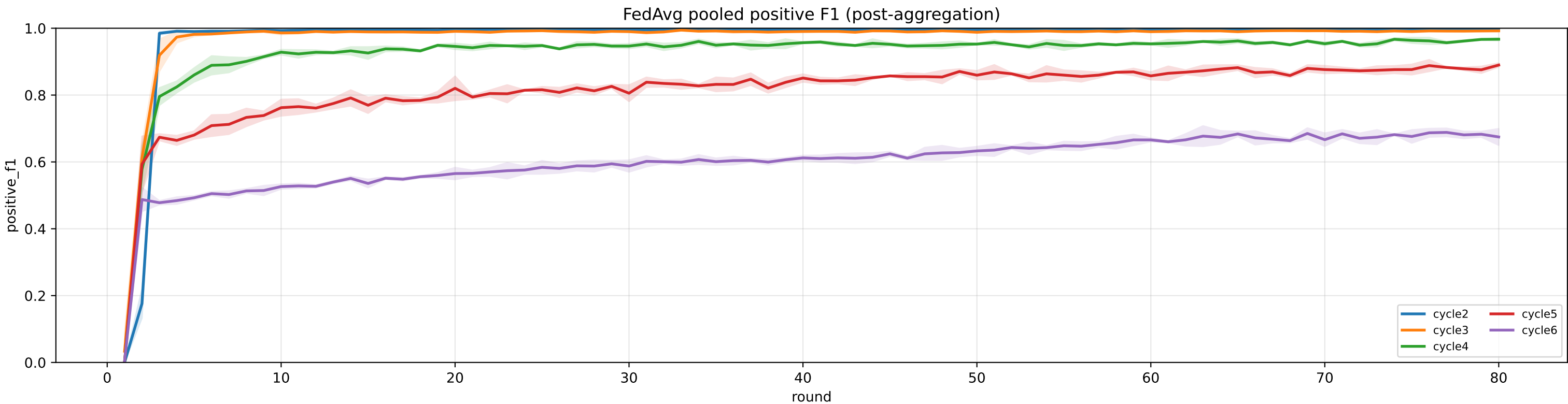
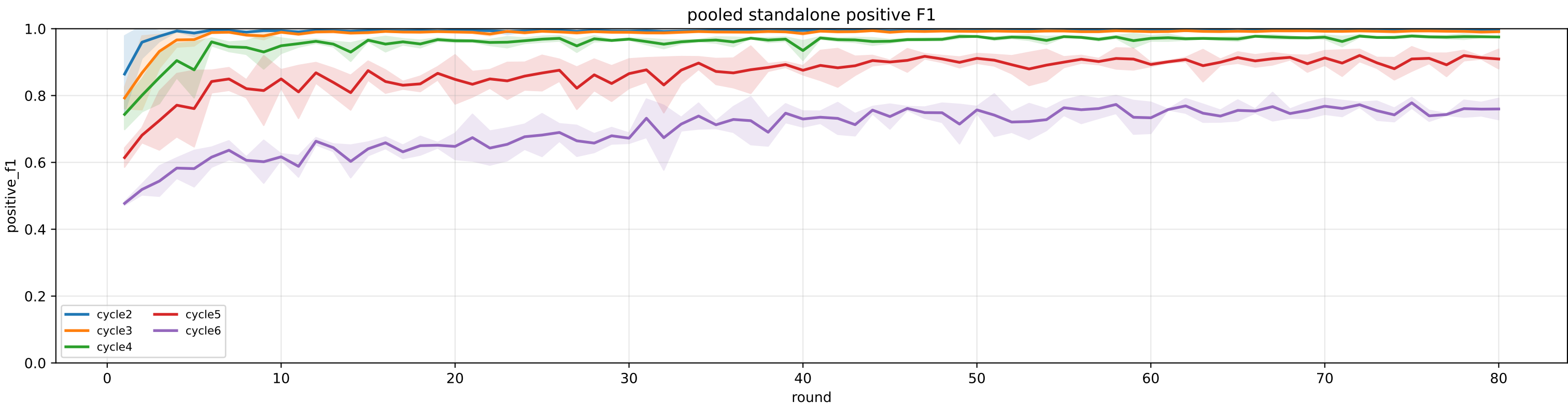
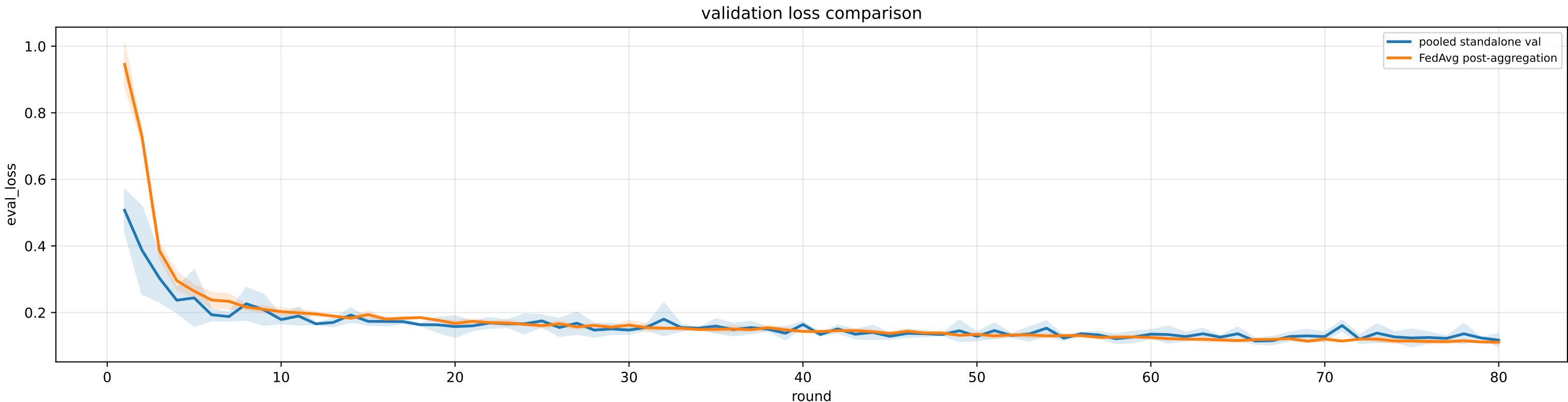
Algorithm	Subset Acc	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	86.7% ± 2.5%	94.5% ± 0.9%	92.8% ± 1.2%	-	-	-	-	-
FedAvg	82.8% ± 0.7%	92.7% ± 0.3%	91.2% ± 0.3%	-	-	-	-	-



graph_id=6 | cycle2=38.3% | cycle3=32.7% | cycle4=27.0% | cycle5=22.8% | cycle6=16.4%

Best test summary (FedAvg vs Local)

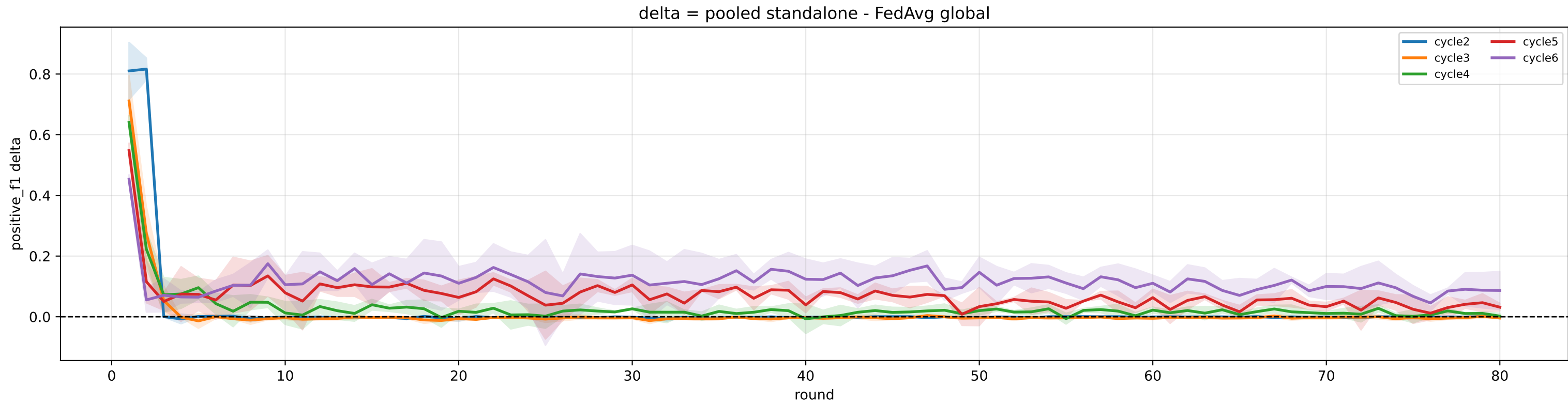
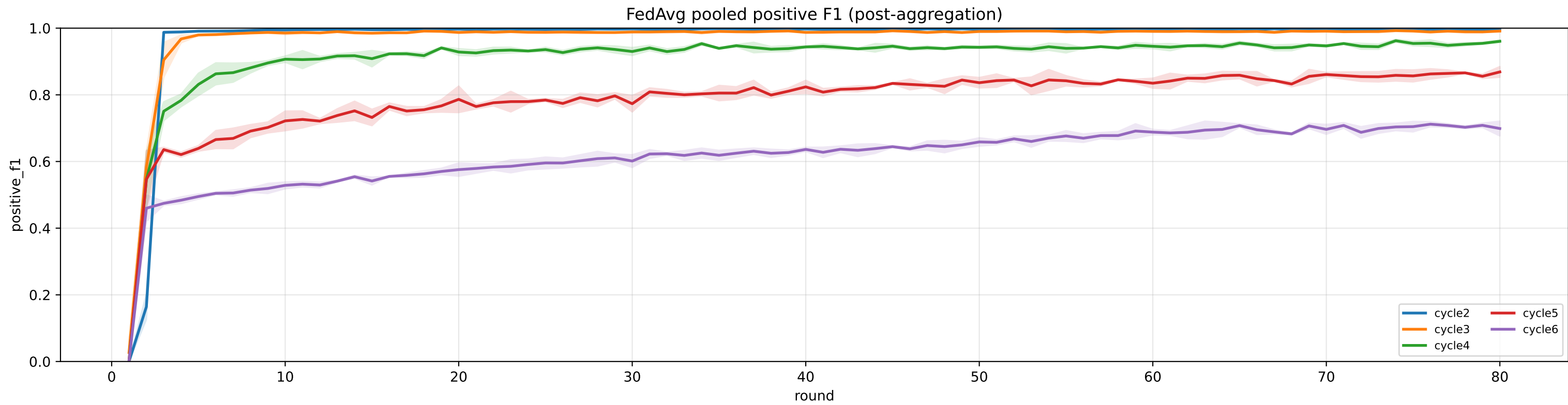
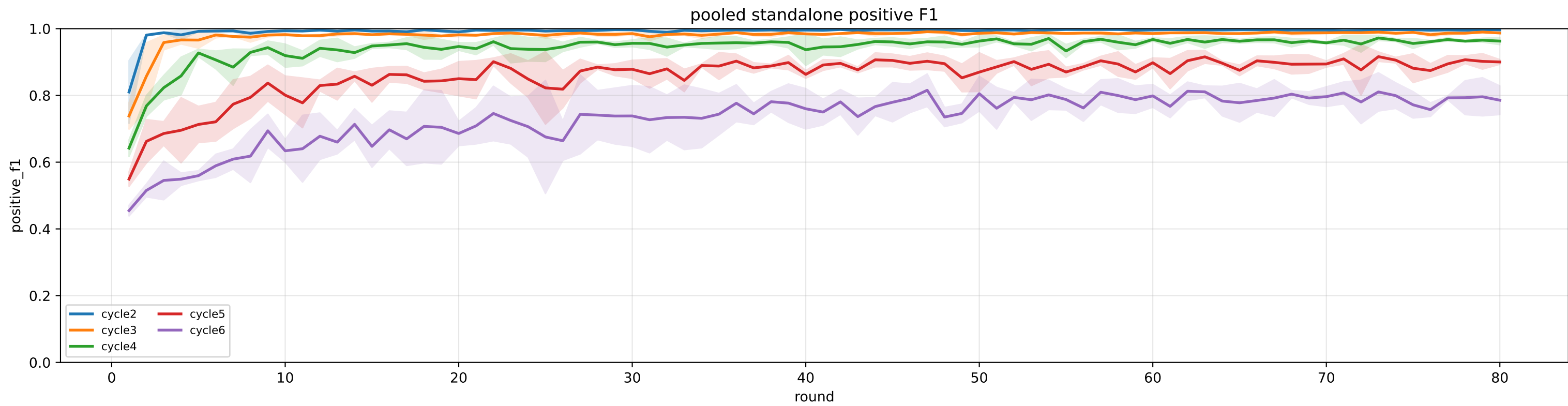
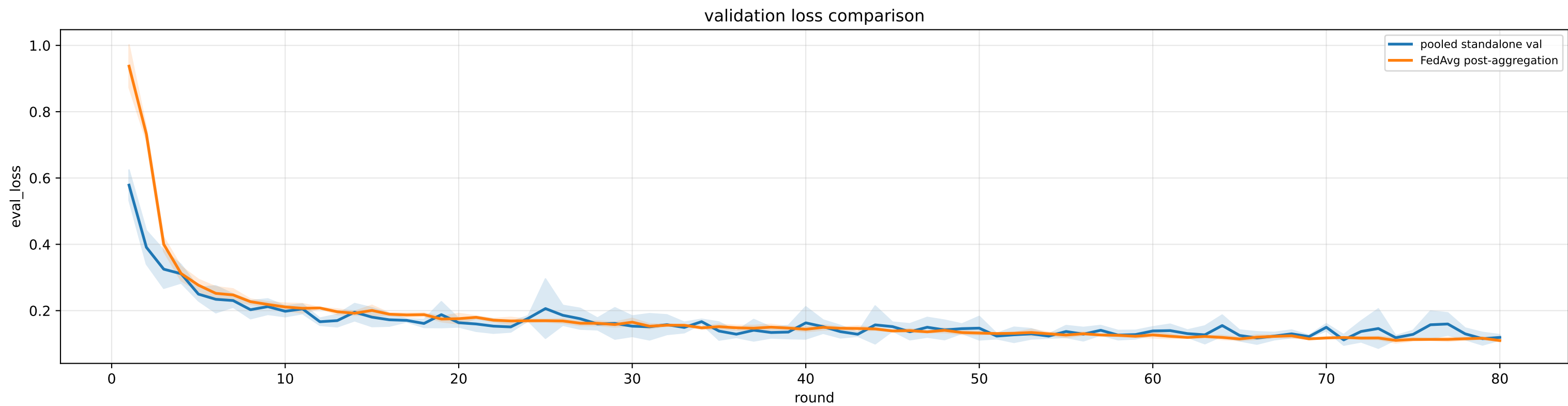
Algorithm	Subset Acc	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	88.9% ± 1.6%	95.0% ± 0.8%	93.5% ± 1.0%	-	-	-	-	-
FedAvg	84.9% ± 0.9%	93.0% ± 0.5%	91.4% ± 0.5%	-	-	-	-	-



graph_id=131 | cycle2=37.4% | cycle3=34.1% | cycle4=28.2% | cycle5=22.2% | cycle6=15.9%

Best test summary (FedAvg vs Local)

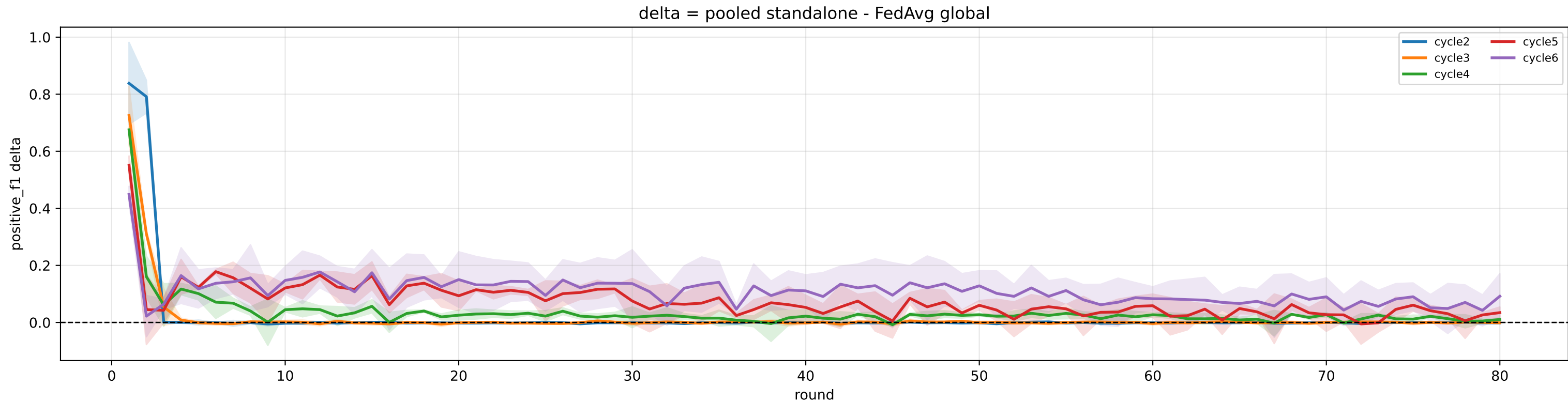
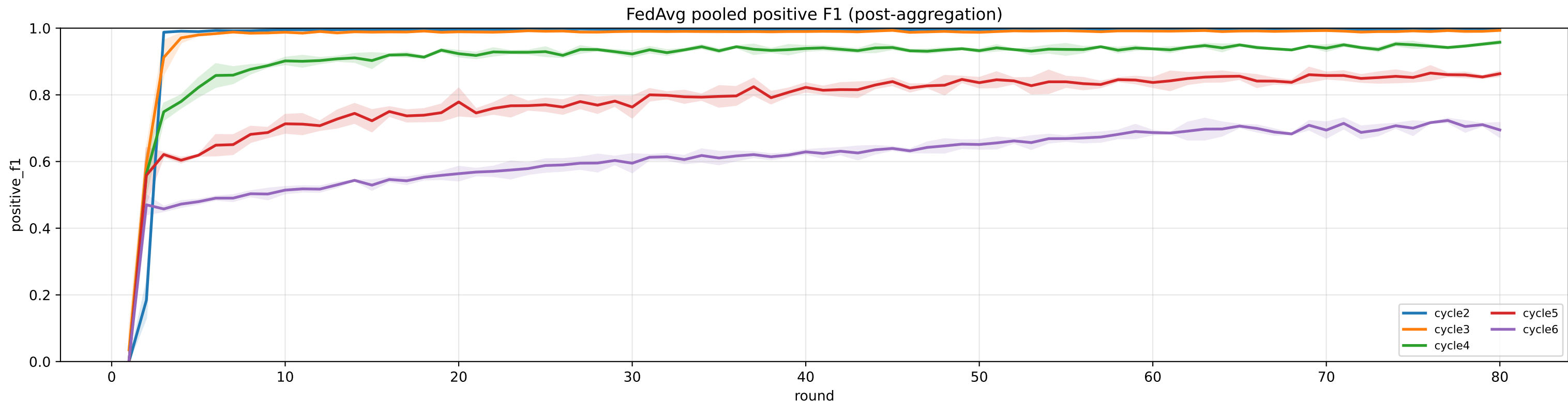
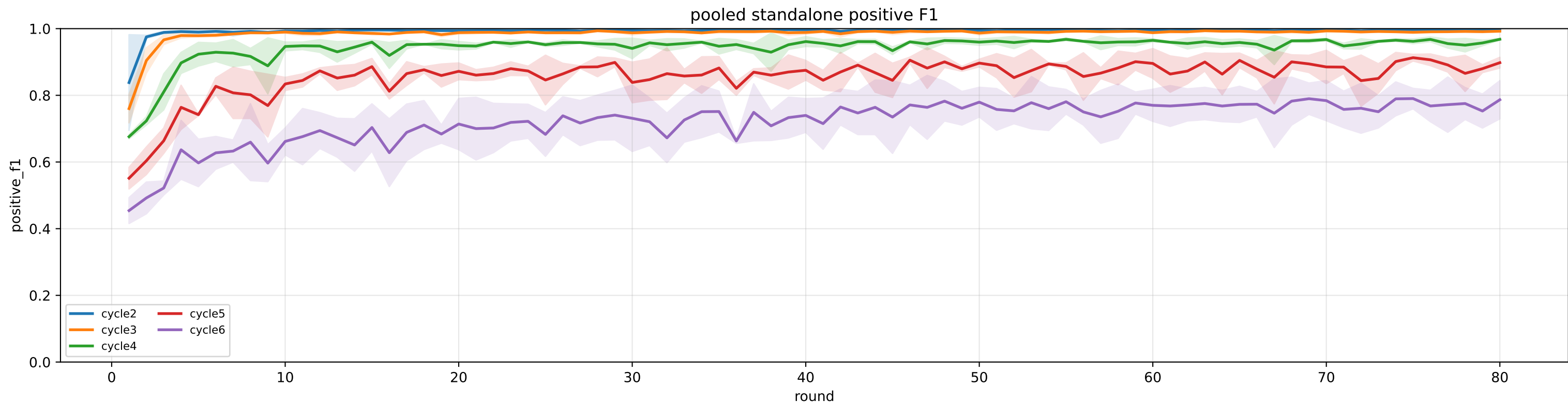
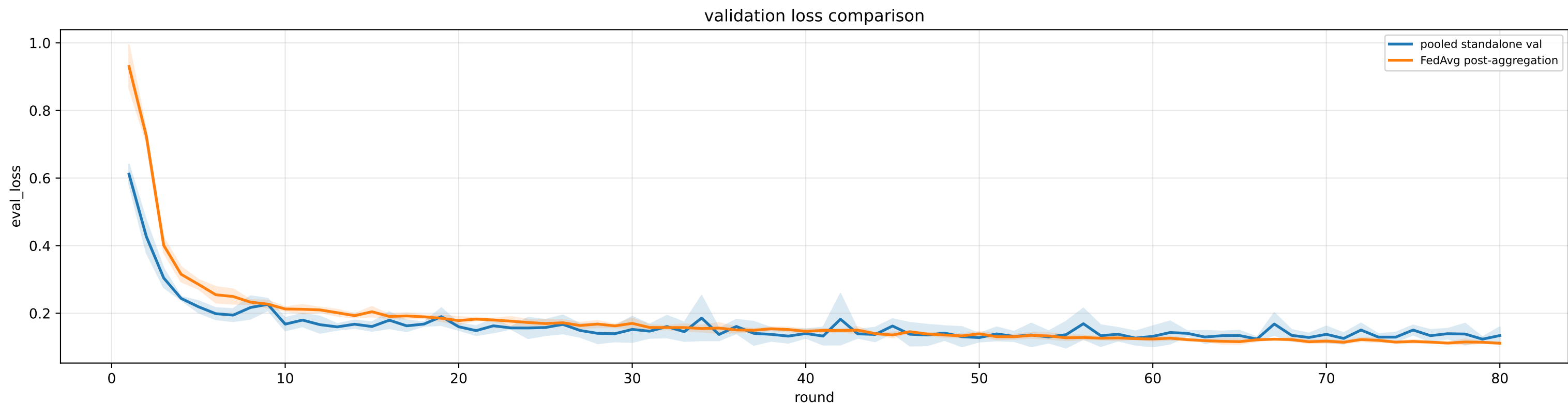
Algorithm	Subset Acc	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	88.4% ± 2.5%	94.6% ± 1.2%	92.4% ± 1.5%	-	-	-	-	-
FedAvg	84.2% ± 0.3%	92.6% ± 0.1%	90.1% ± 0.1%	-	-	-	-	-



graph_id=119 | cycle2=38.0% | cycle3=32.3% | cycle4=28.5% | cycle5=23.0% | cycle6=16.1%

Best test summary (FedAvg vs Local)

Algorithm	Subset Acc	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	85.9% ± 1.1%	93.7% ± 0.5%	91.2% ± 0.7%	-	-	-	-	-
FedAvg	83.2% ± 0.9%	92.2% ± 0.4%	89.8% ± 0.4%	-	-	-	-	-

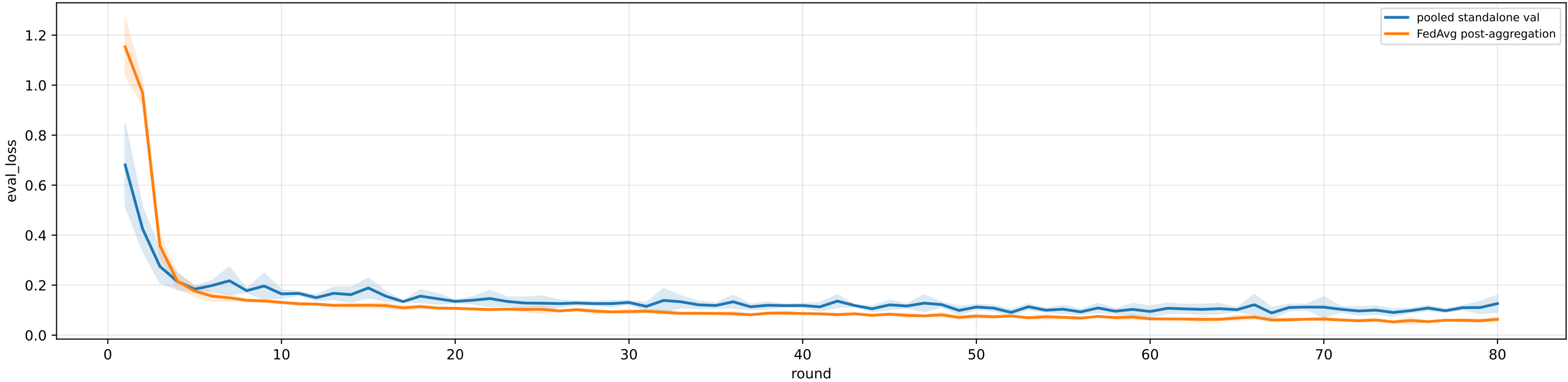


graph_id=143 | cycle2=18.0% | cycle3=15.2% | cycle4=13.5% | cycle5=10.7% | cycle6=7.9%

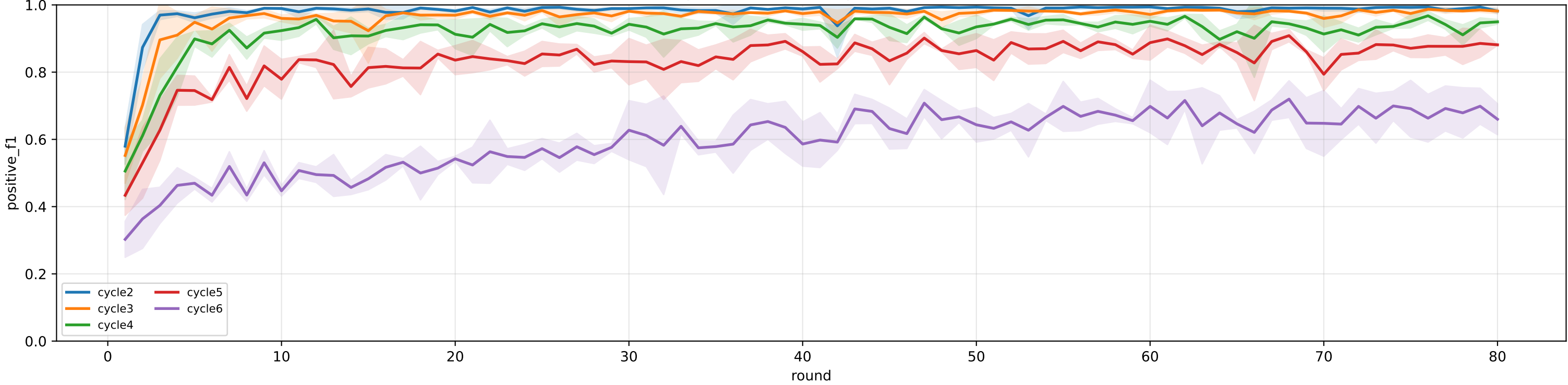
Best test summary (FedAvg vs Local)

Algorithm	Subset Acc	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	90.7% ± 1.4%	91.4% ± 1.2%	88.9% ± 1.3%	-	-	-	-	-
FedAvg	94.6% ± 0.2%	94.9% ± 0.3%	92.9% ± 0.3%	-	-	-	-	-

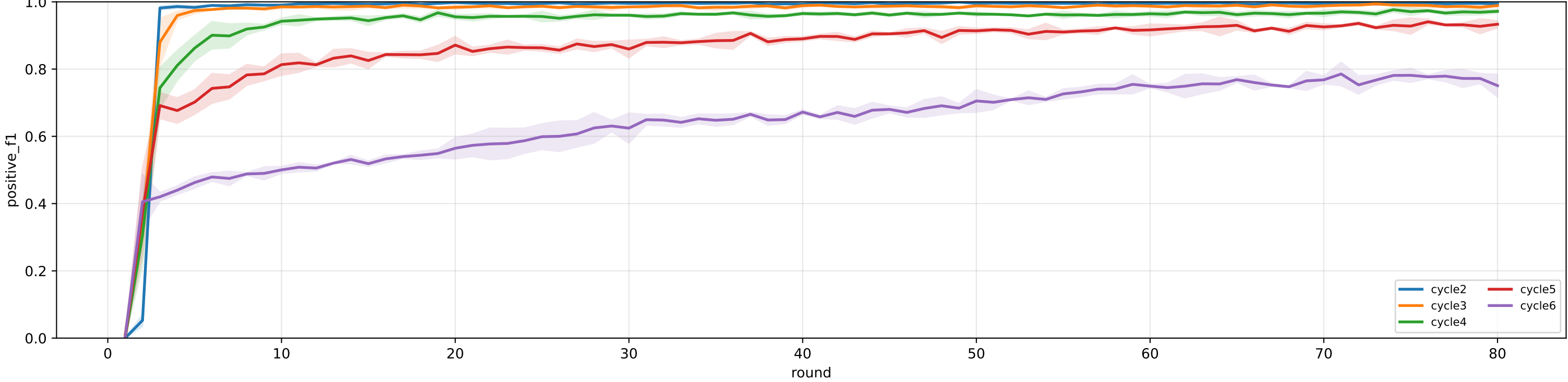
validation loss comparison



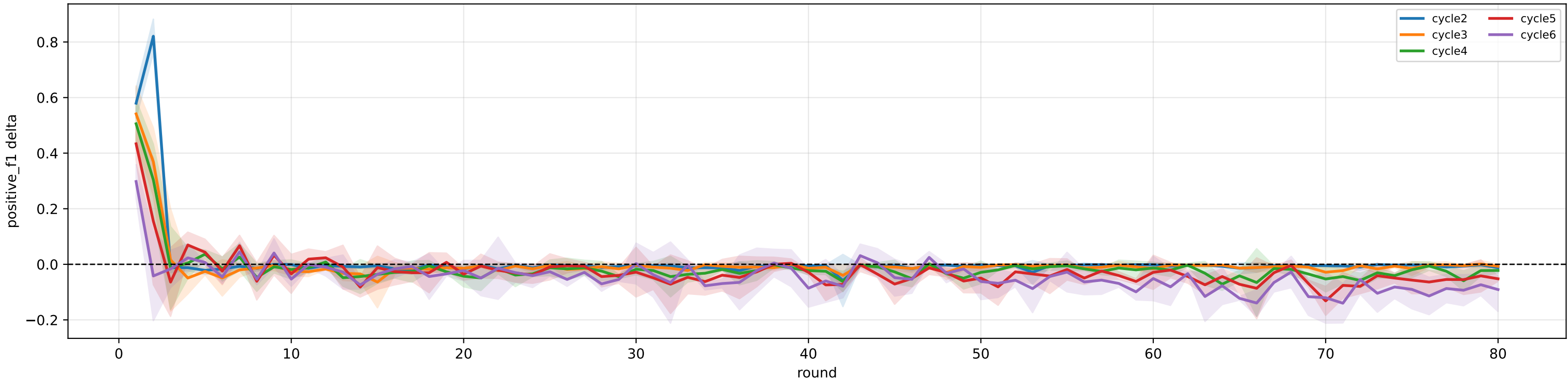
pooled standalone positive F1



FedAvg pooled positive F1 (post-aggregation)



delta = pooled standalone - FedAvg global

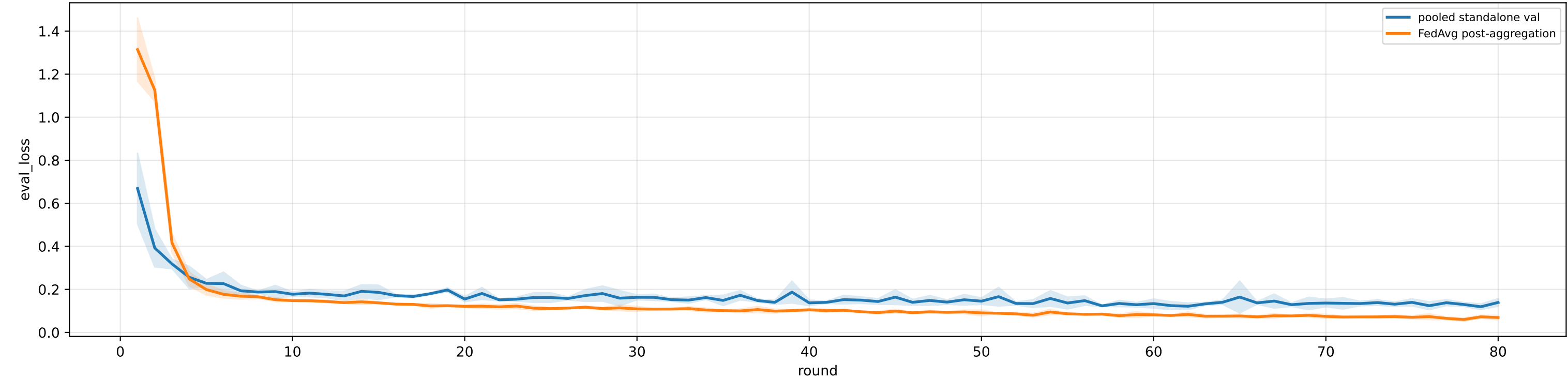


graph_id=114 | cycle2=17.6% | cycle3=14.3% | cycle4=12.5% | cycle5=9.1% | cycle6=7.6%

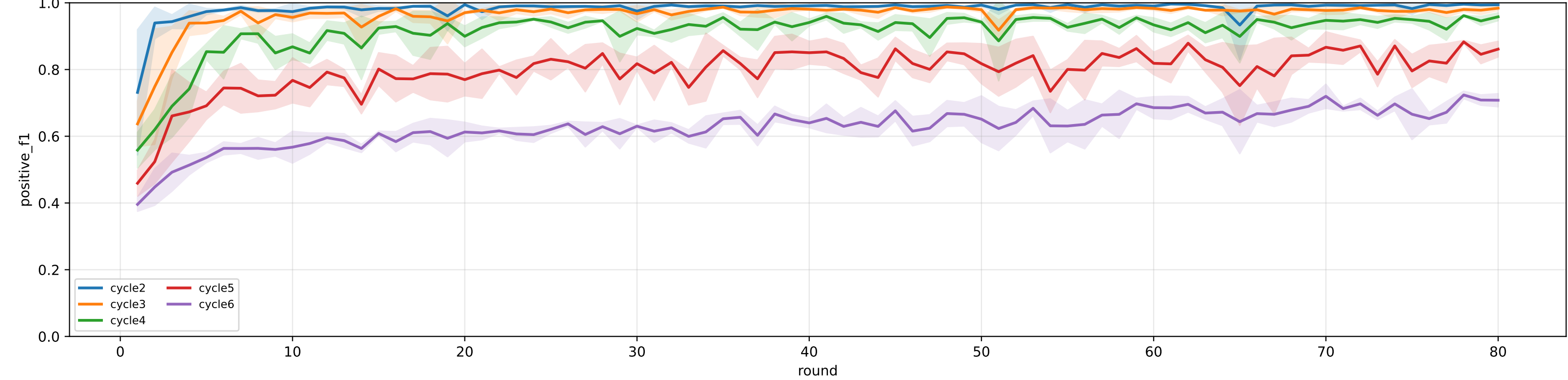
Best test summary (FedAvg vs Local)

Algorithm	Subset Acc	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	91.0% ± 0.7%	91.3% ± 0.5%	88.7% ± 0.5%	-	-	-	-	-
FedAvg	94.8% ± 0.4%	94.9% ± 0.4%	92.9% ± 0.4%	-	-	-	-	-

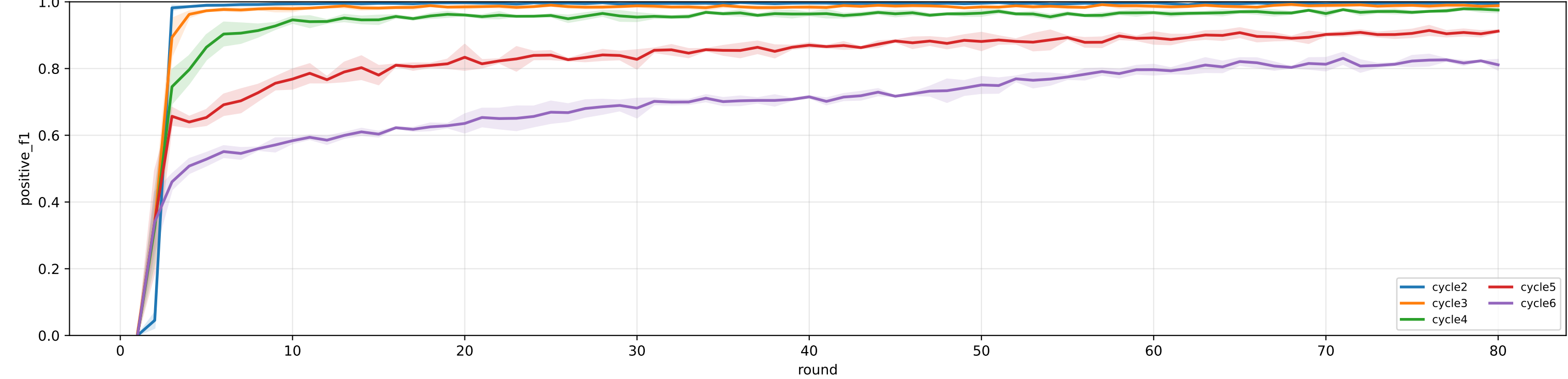
validation loss comparison



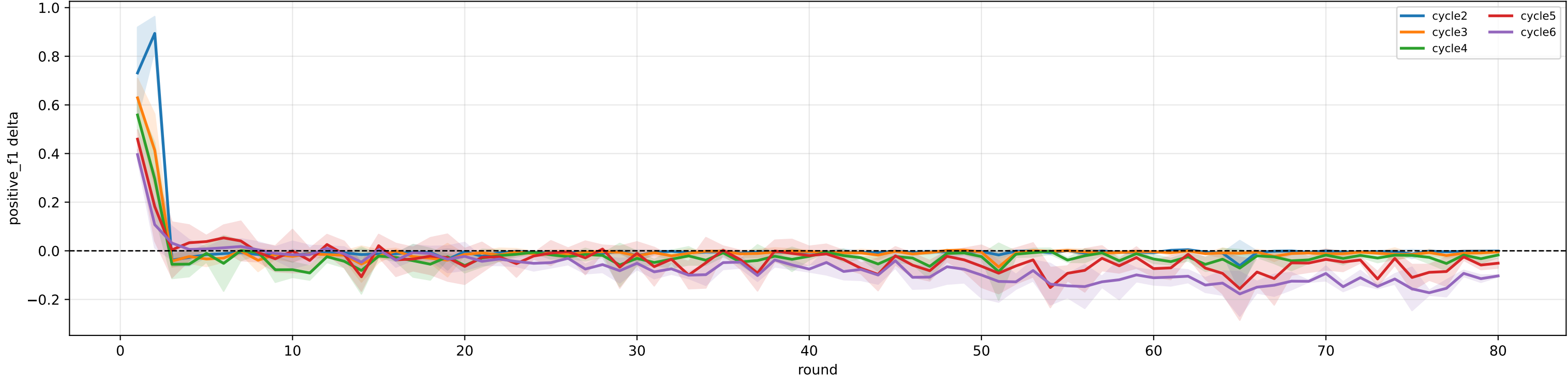
pooled standalone positive F1



FedAvg pooled positive F1 (post-aggregation)



delta = pooled standalone - FedAvg global

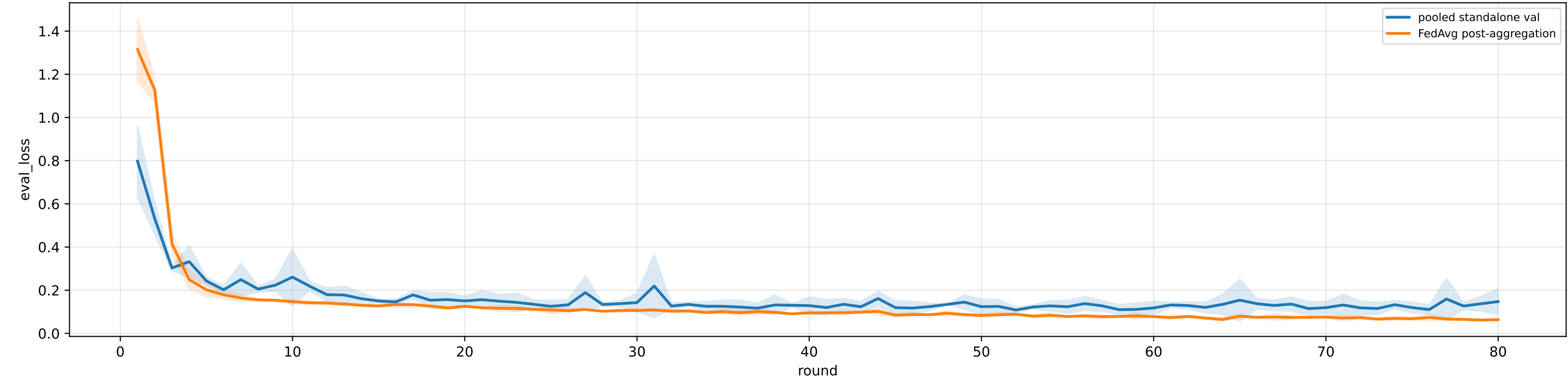


graph_id=117 | cycle2=16.7% | cycle3=15.9% | cycle4=12.2% | cycle5=8.5% | cycle6=7.2%

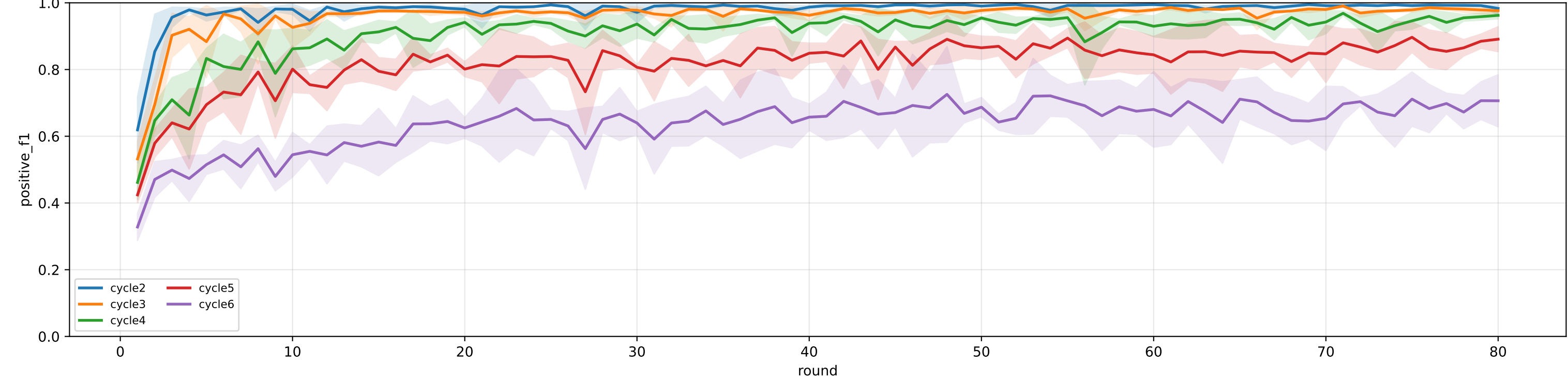
Best test summary (FedAvg vs Local)

Algorithm	Subset Acc	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	90.4% ± 2.2%	90.6% ± 2.2%	89.5% ± 2.0%	-	-	-	-	-
FedAvg	94.8% ± 0.2%	94.8% ± 0.2%	93.8% ± 0.2%	-	-	-	-	-

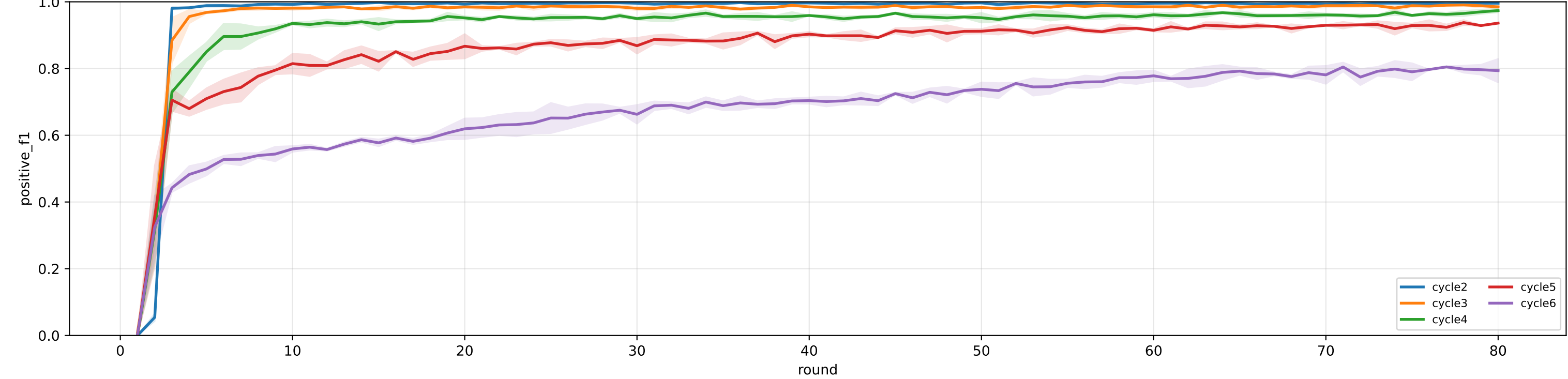
validation loss comparison



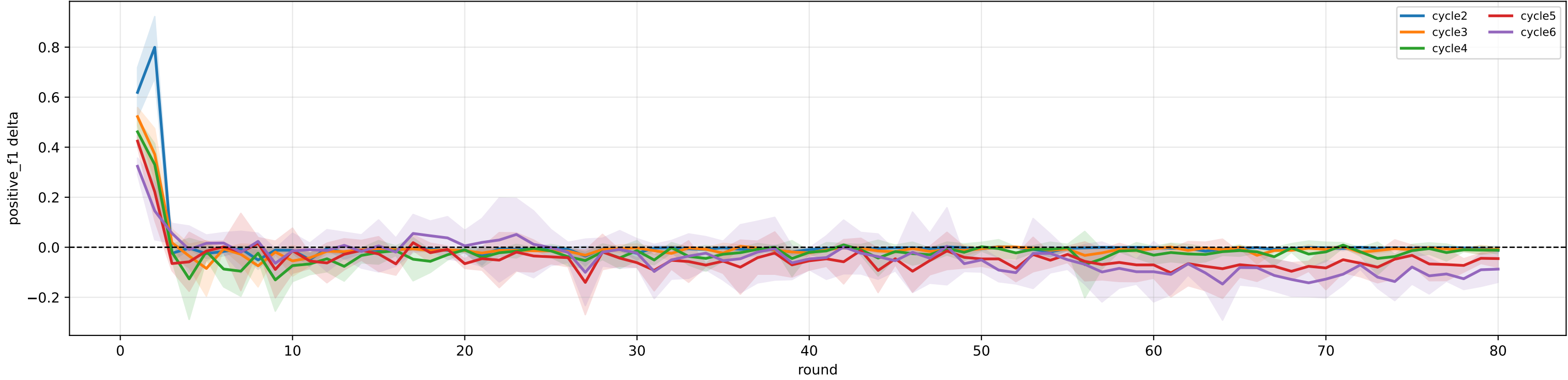
pooled standalone positive F1



FedAvg pooled positive F1 (post-aggregation)



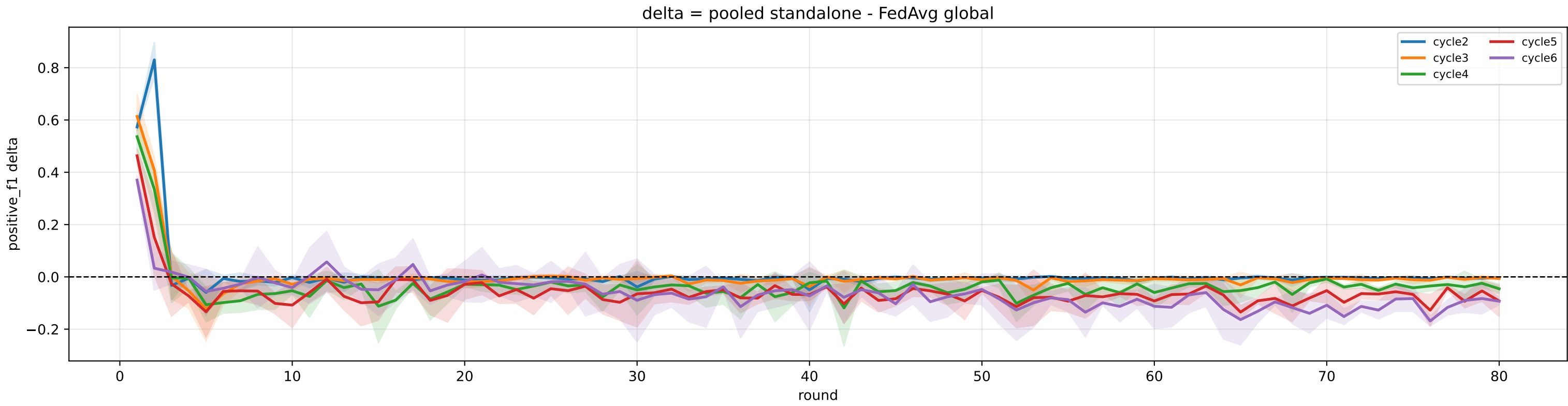
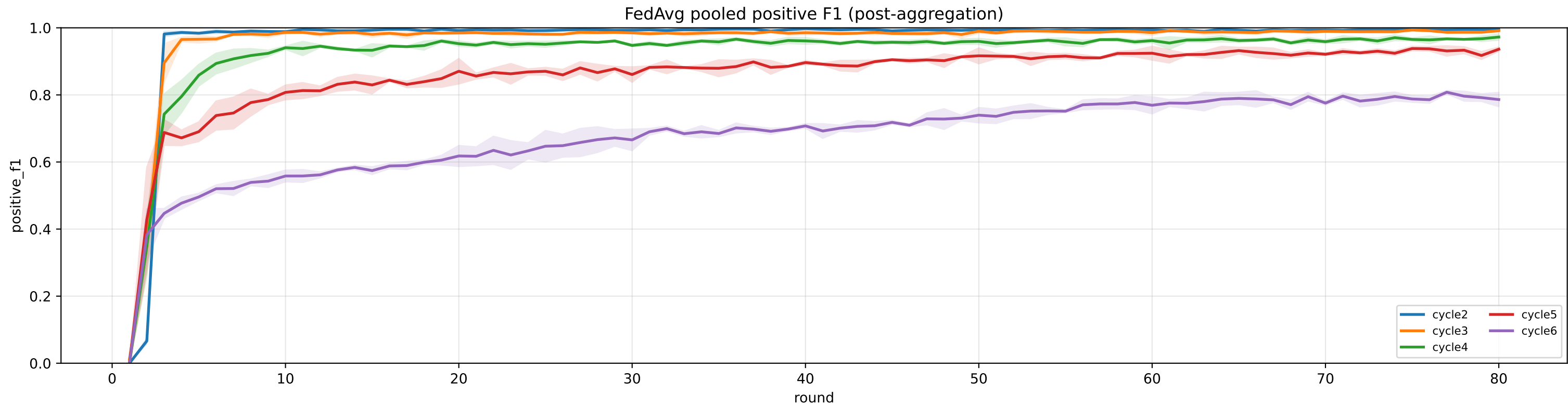
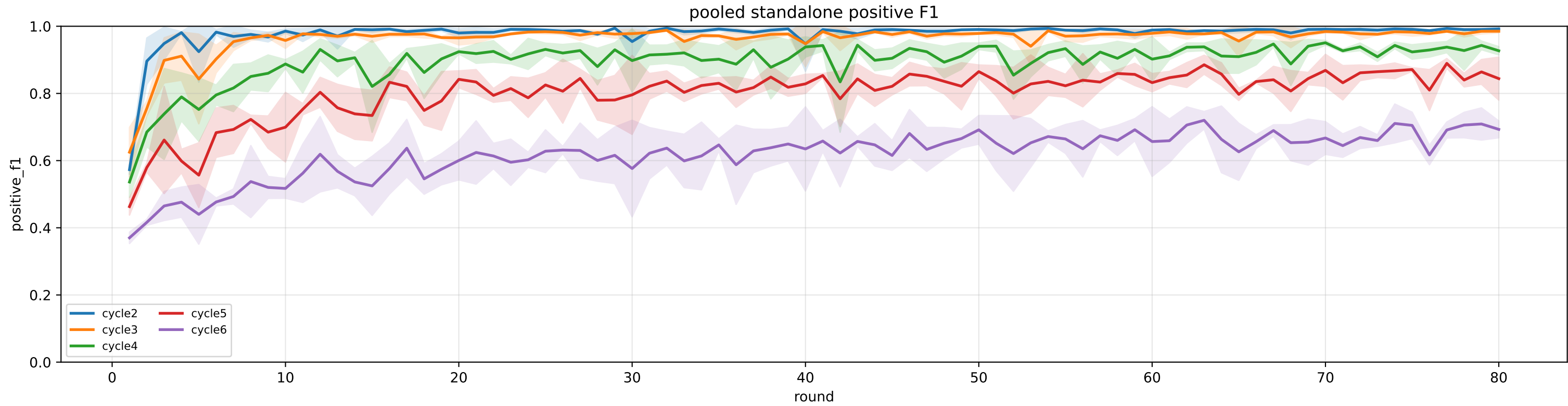
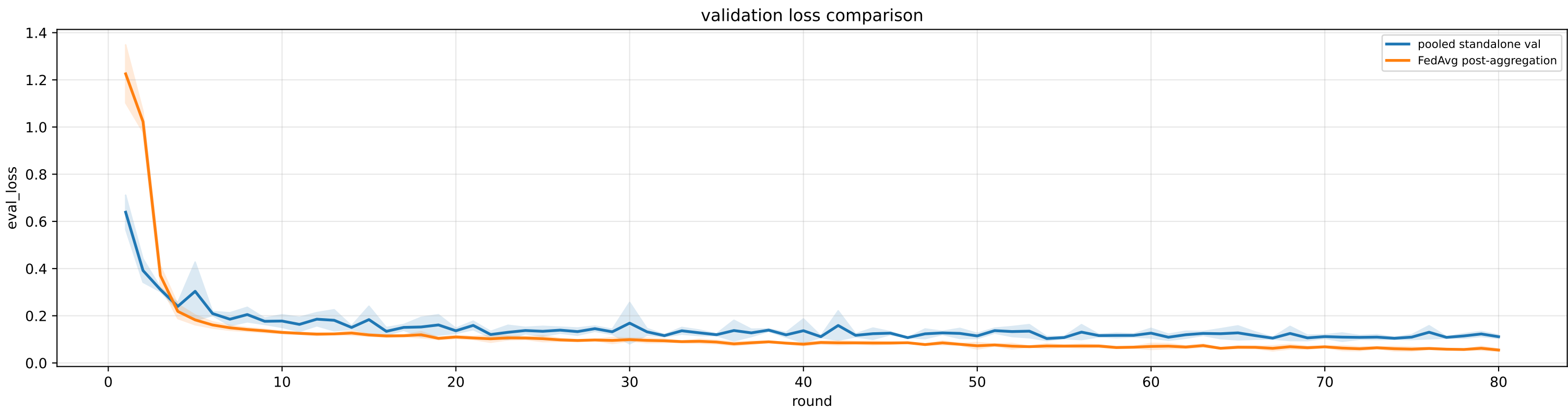
delta = pooled standalone - FedAvg global



graph_id=2 | cycle2=16.3% | cycle3=13.0% | cycle4=12.2% | cycle5=9.7% | cycle6=8.2%

Best test summary (FedAvg vs Local)

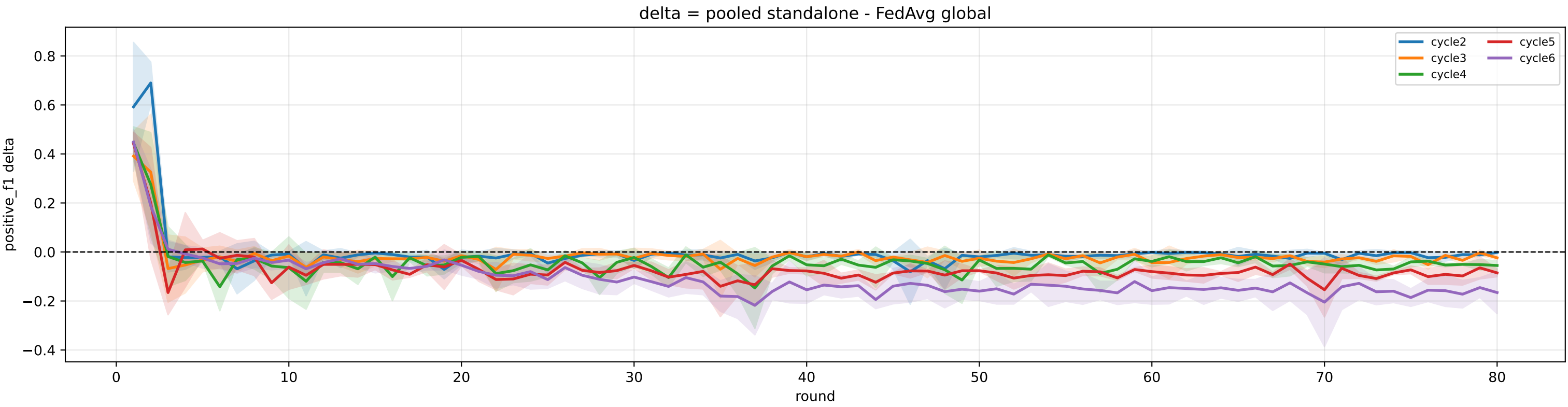
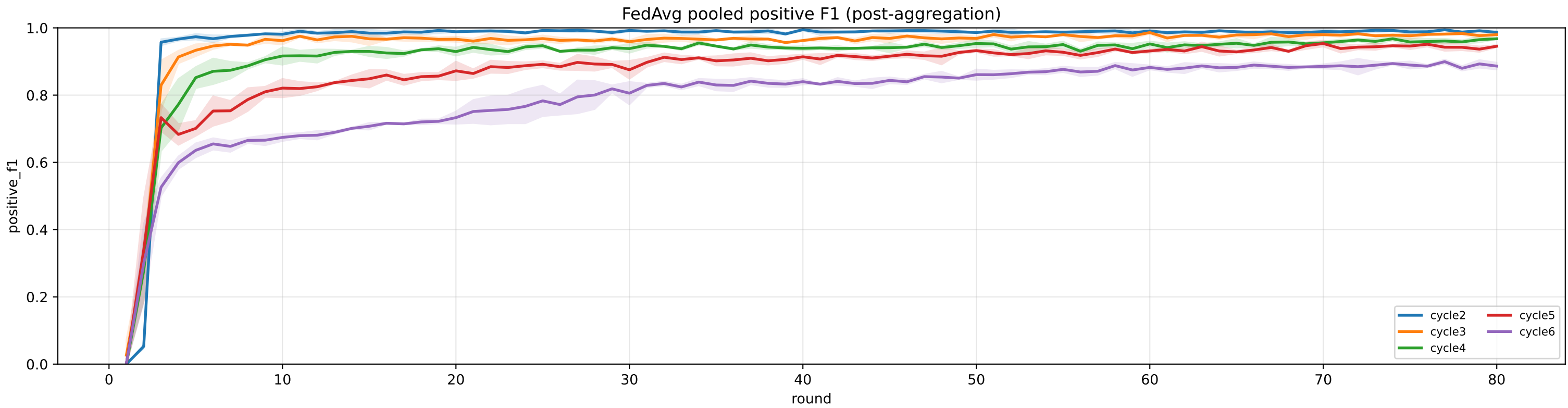
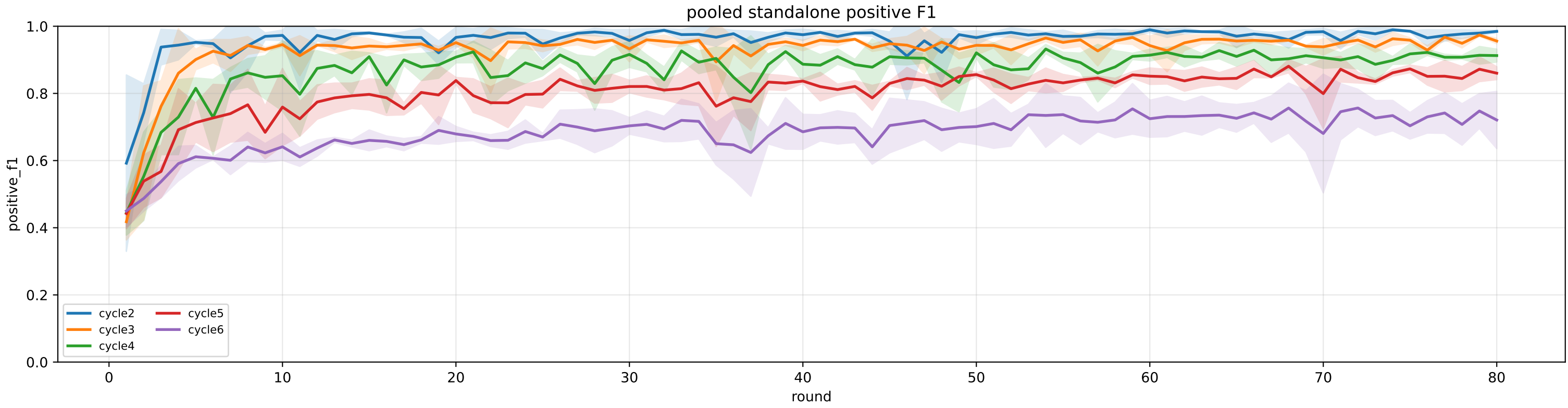
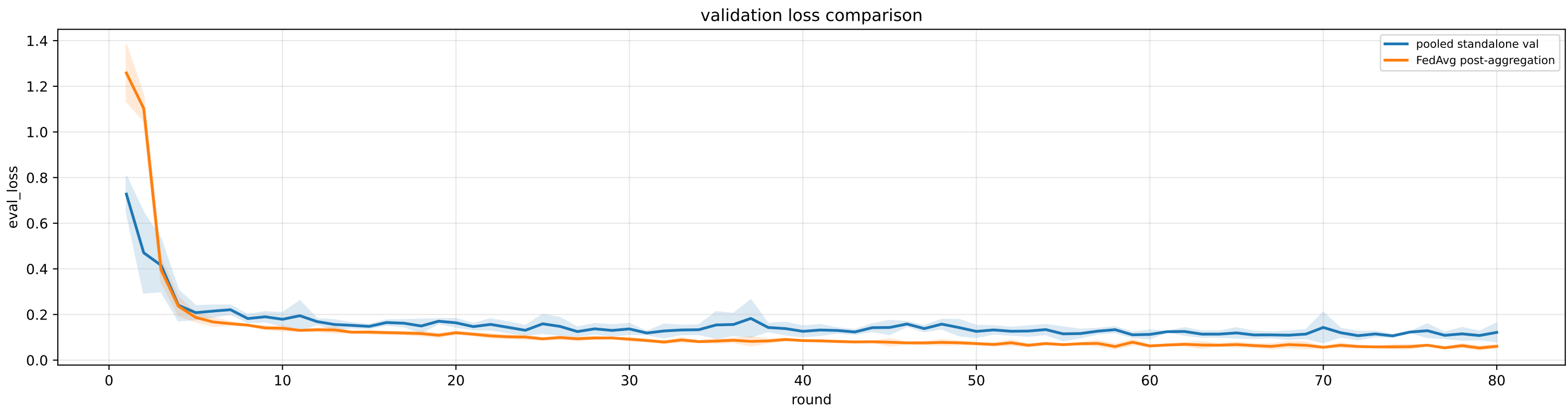
Algorithm	Subset Acc	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	90.4% ± 1.3%	91.1% ± 1.2%	90.2% ± 1.2%	-	-	-	-	-
FedAvg	95.0% ± 0.4%	95.4% ± 0.4%	94.5% ± 0.4%	-	-	-	-	-



graph_id=87 | cycle2=8.4% | cycle3=11.0% | cycle4=10.2% | cycle5=9.4% | cycle6=9.2%

Best test summary (FedAvg vs Local)

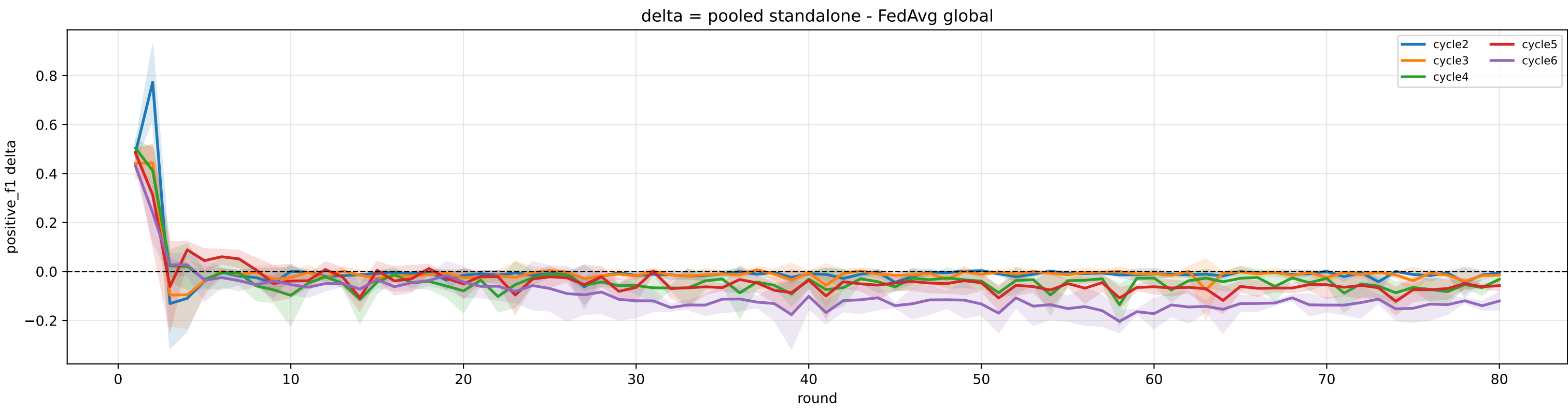
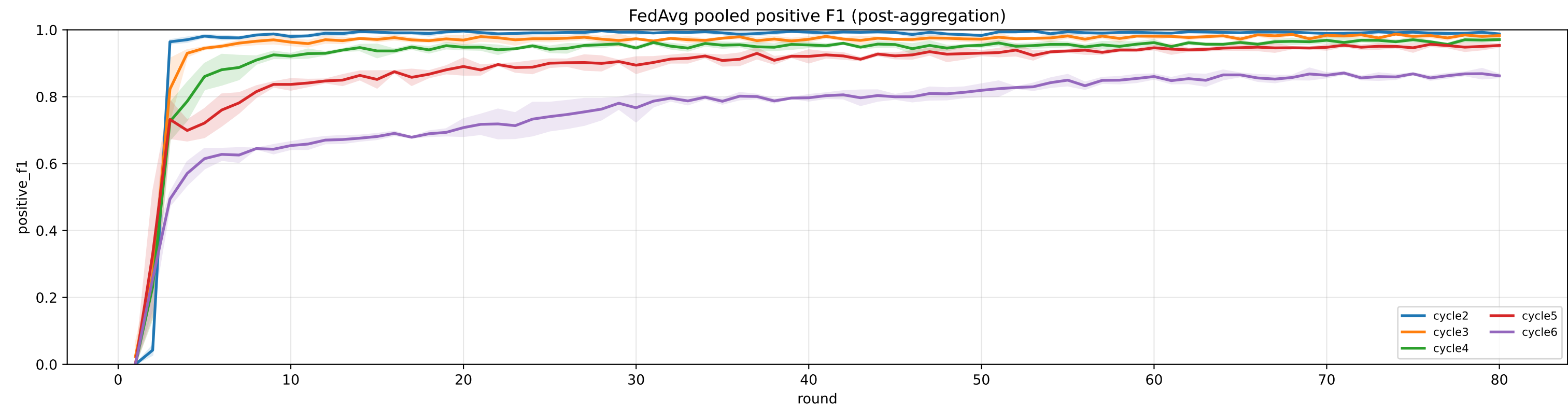
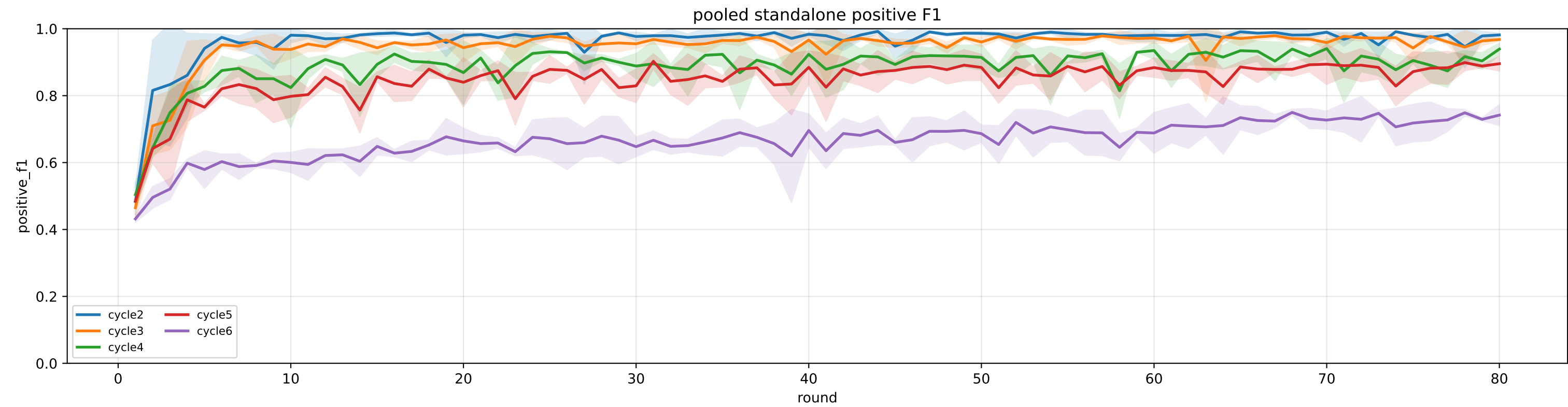
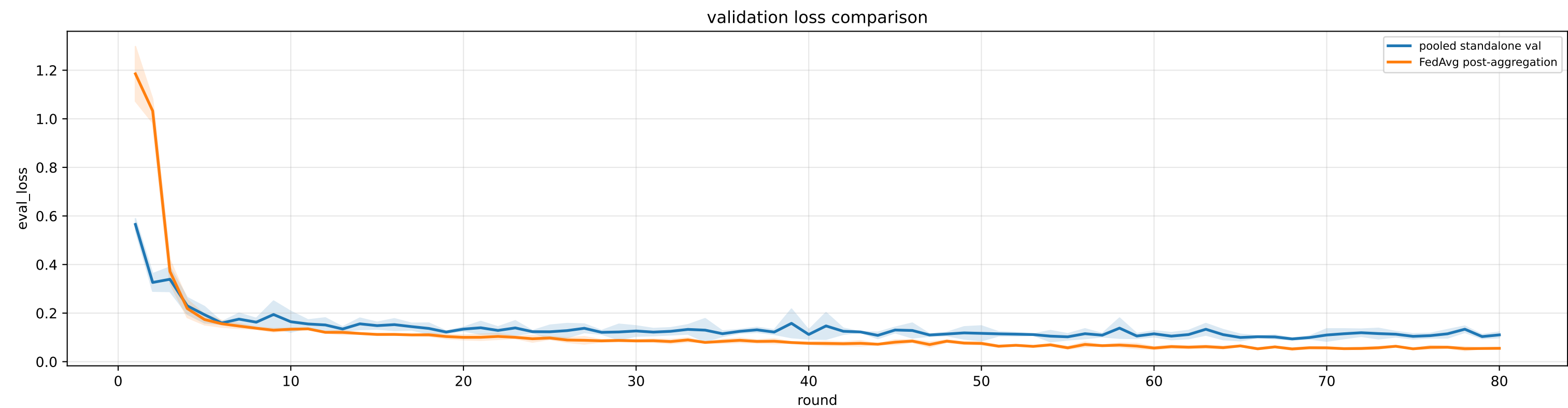
Algorithm	Subset Acc	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	88.7% ± 1.1%	86.3% ± 0.8%	87.4% ± 0.6%	-	-	-	-	-
FedAvg	95.4% ± 0.3%	94.4% ± 0.6%	94.4% ± 0.6%	-	-	-	-	-



graph_id=157 | cycle2=9.3% | cycle3=10.7% | cycle4=10.8% | cycle5=12.3% | cycle6=9.3%

Best test summary (FedAvg vs Local)

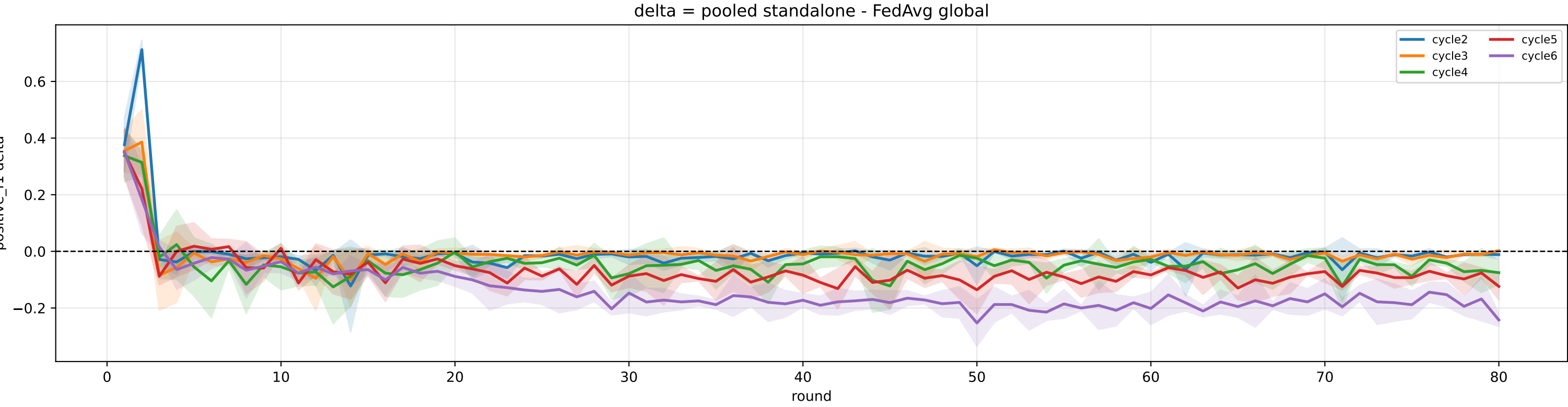
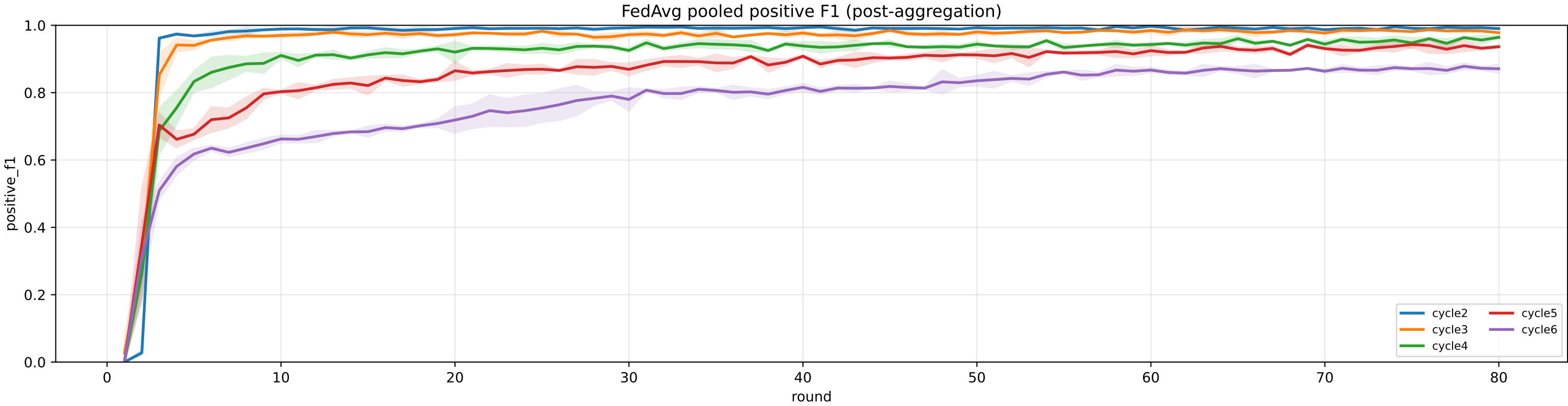
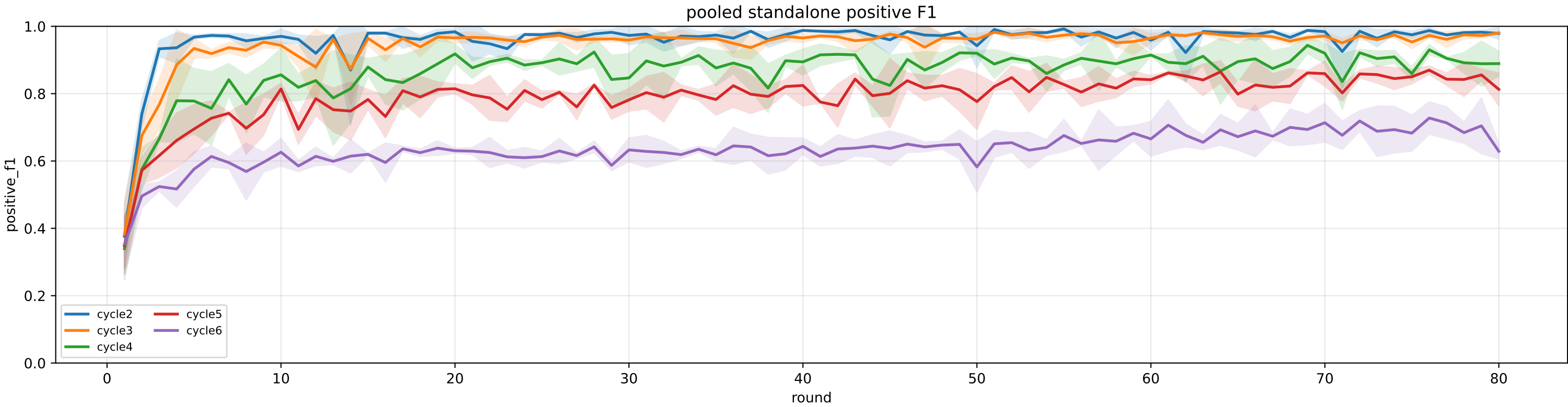
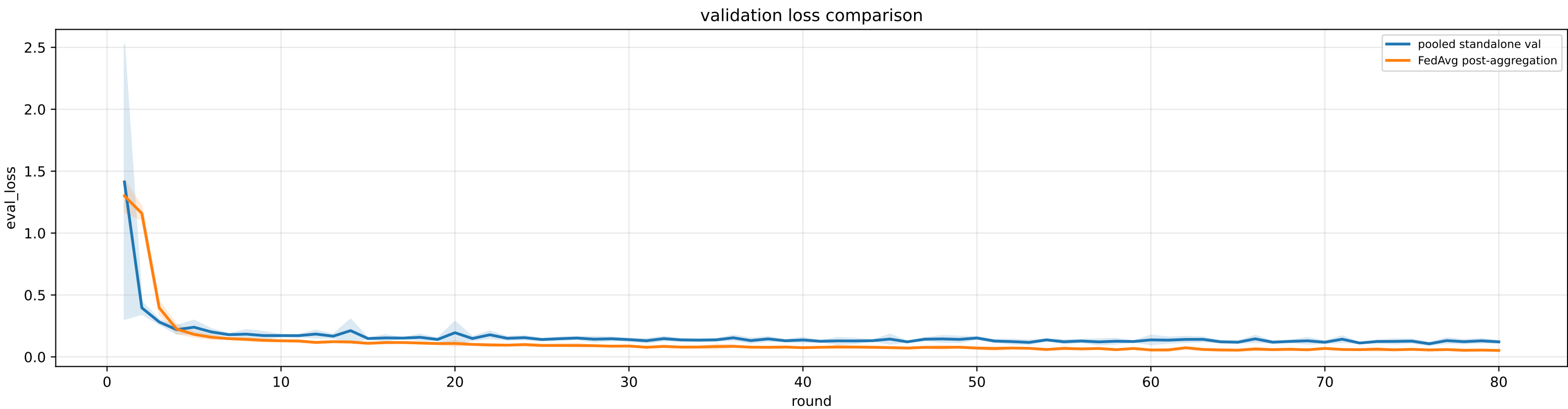
Algorithm	Subset Acc	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	89.9% ± 0.9%	88.2% ± 1.1%	89.1% ± 0.8%	-	-	-	-	-
FedAvg	95.6% ± 0.2%	94.9% ± 0.2%	94.9% ± 0.2%	-	-	-	-	-



graph_id=149 | cycle2=9.2% | cycle3=8.9% | cycle4=10.8% | cycle5=7.9% | cycle6=8.6%

Best test summary (FedAvg vs Local)

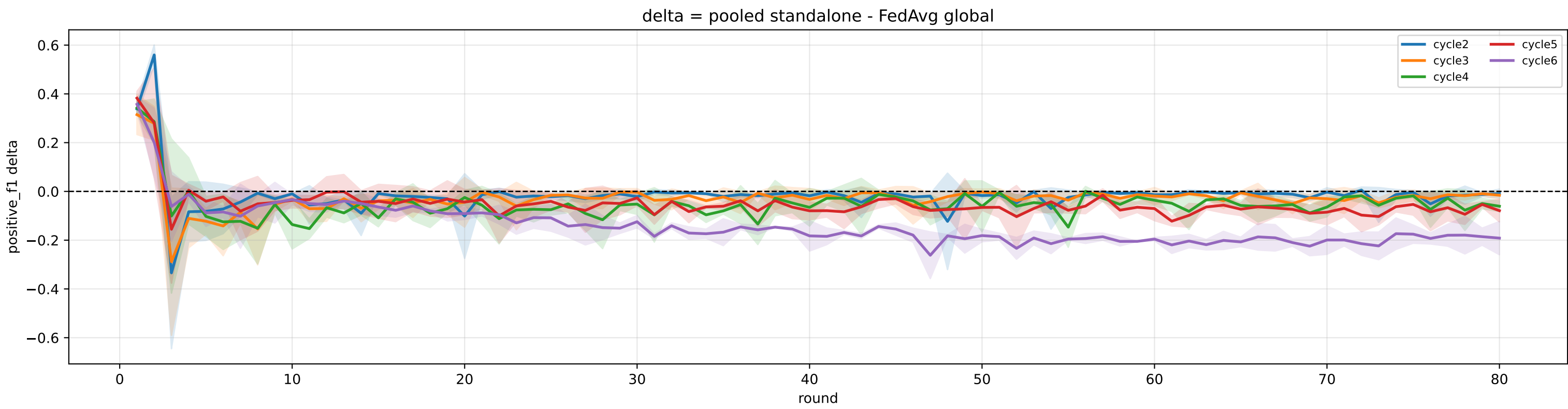
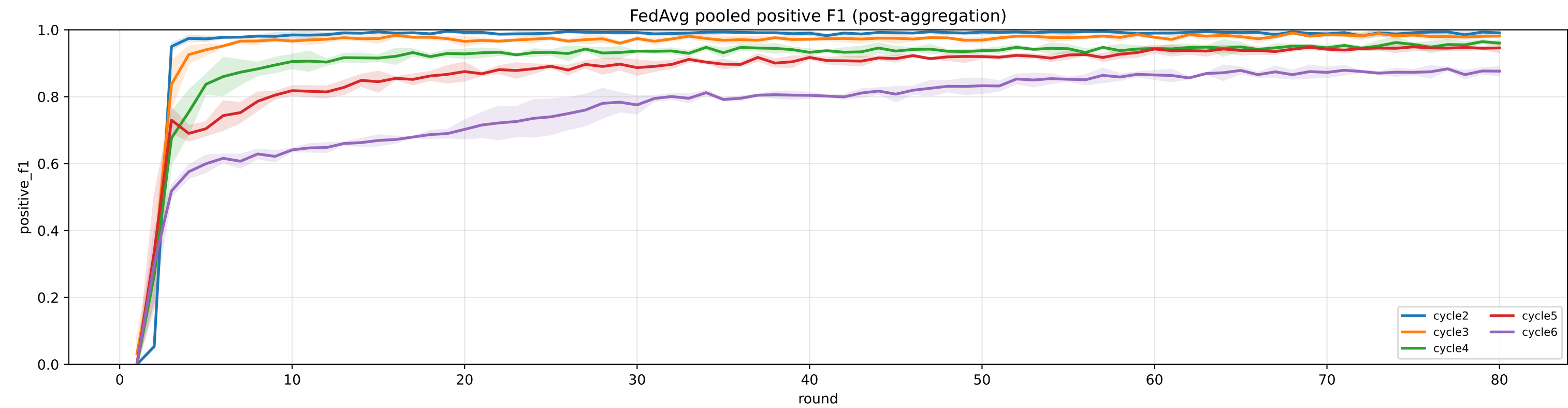
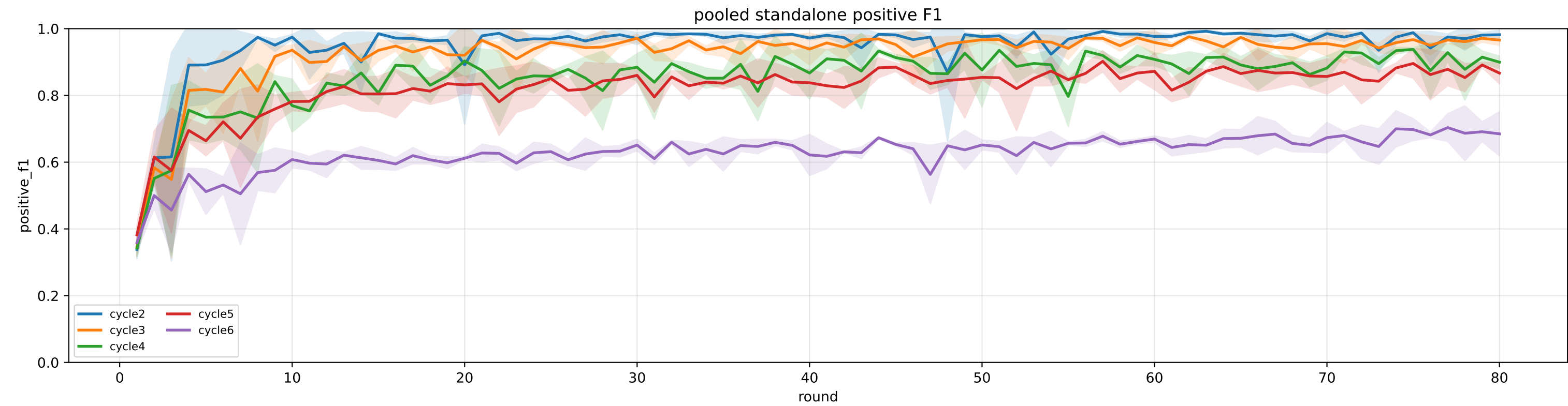
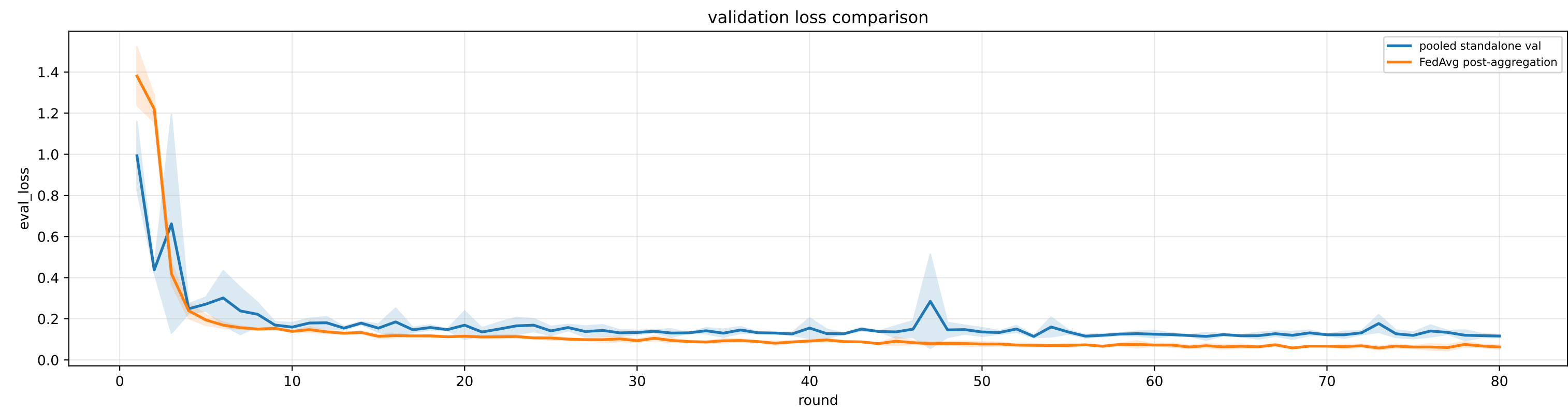
Algorithm	Subset Acc	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	89.6% ± 1.7%	87.8% ± 1.5%	89.2% ± 0.9%	-	-	-	-	-
FedAvg	96.3% ± 0.2%	95.3% ± 0.4%	95.4% ± 0.4%	-	-	-	-	-



graph_id=109 | cycle2=7.4% | cycle3=10.0% | cycle4=7.1% | cycle5=7.6% | cycle6=7.4%

Best test summary (FedAvg vs Local)

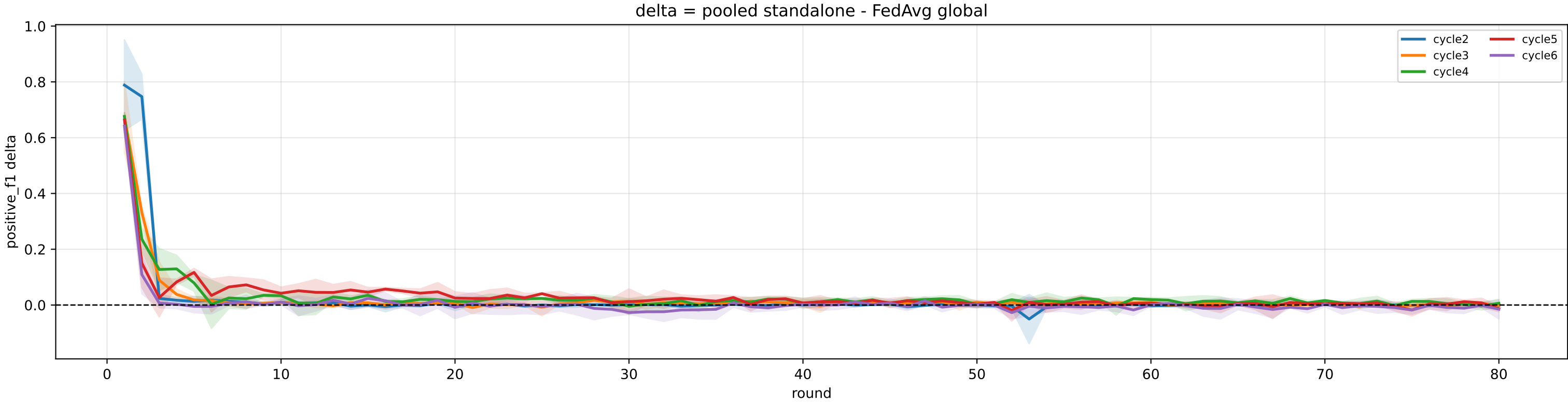
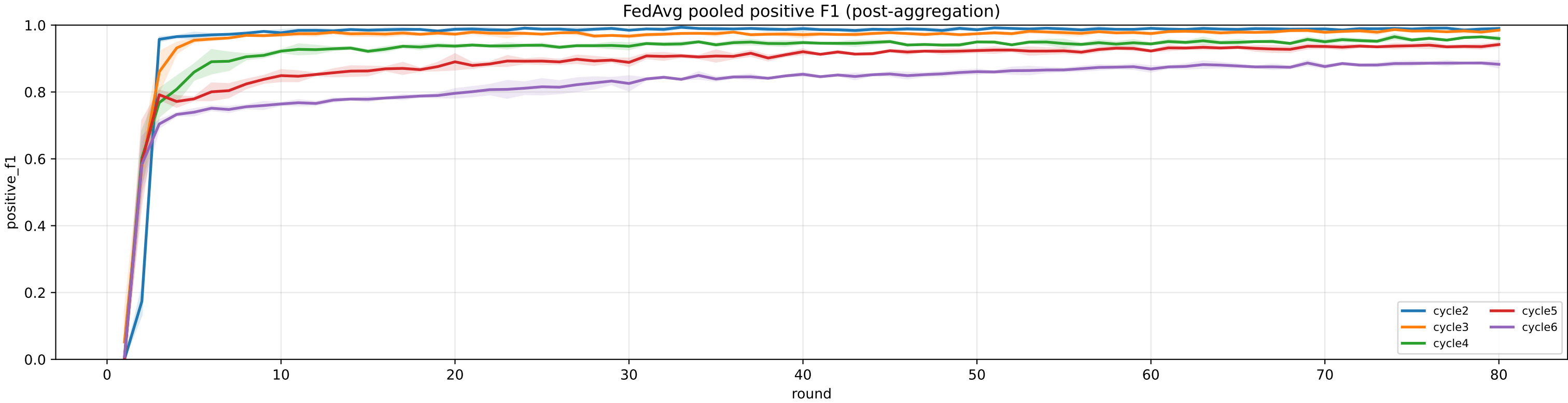
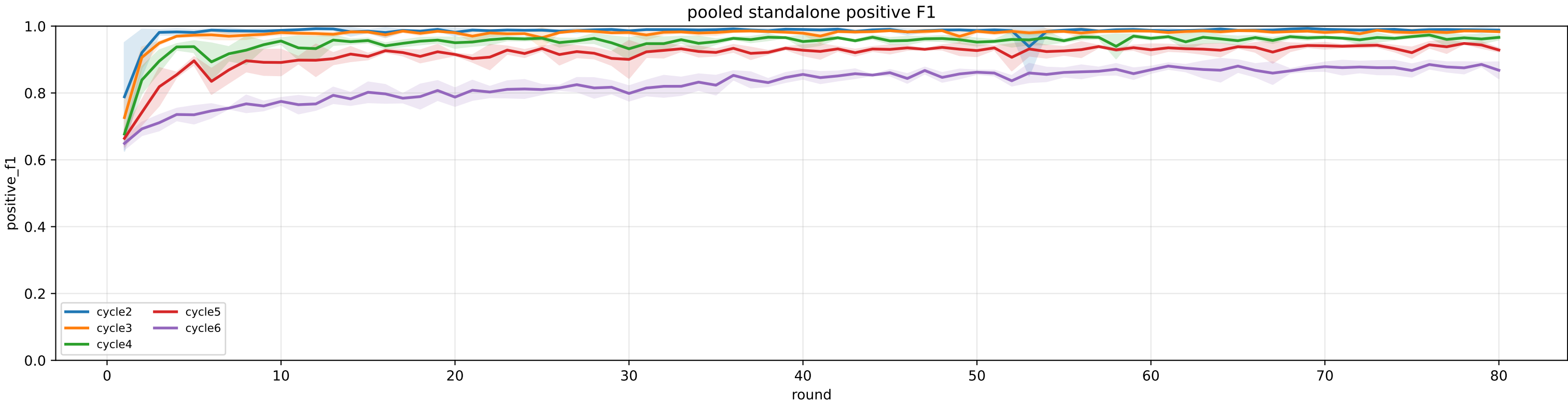
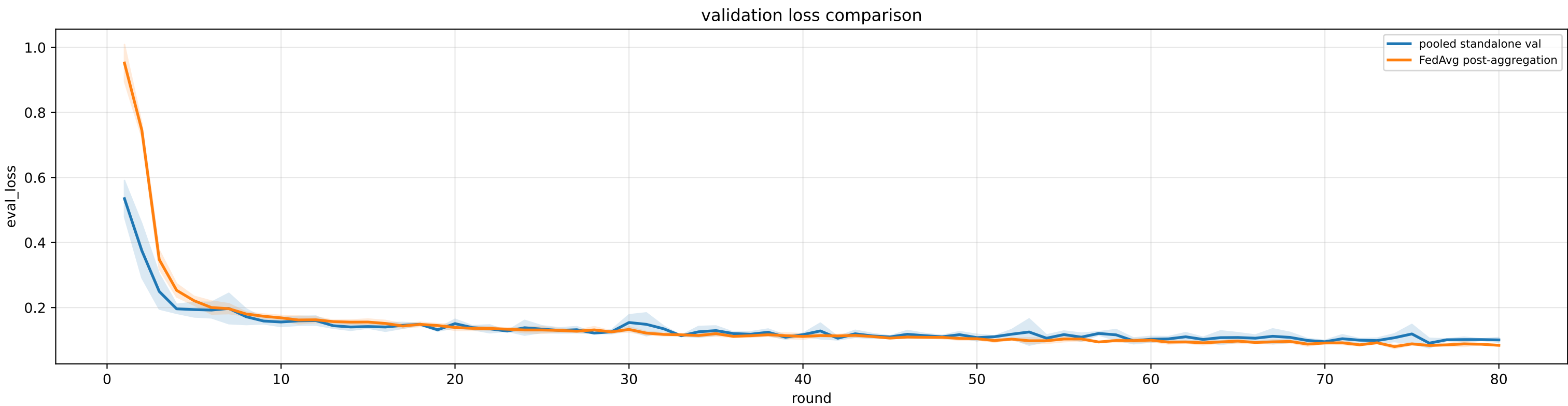
Algorithm	Subset Acc	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	90.7% ± 1.2%	88.0% ± 1.0%	89.8% ± 0.5%	-	-	-	-	-
FedAvg	96.0% ± 0.3%	94.8% ± 0.5%	95.2% ± 0.5%	-	-	-	-	-



graph_id=88 | cycle2=20.0% | cycle3=24.4% | cycle4=27.4% | cycle5=29.0% | cycle6=28.5%

Best test summary (FedAvg vs Local)

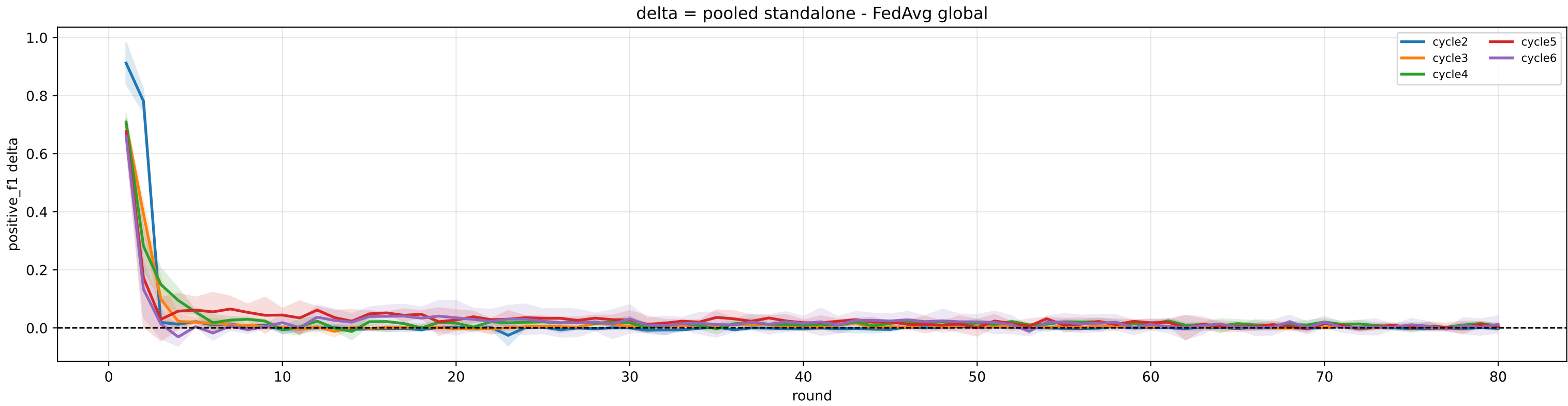
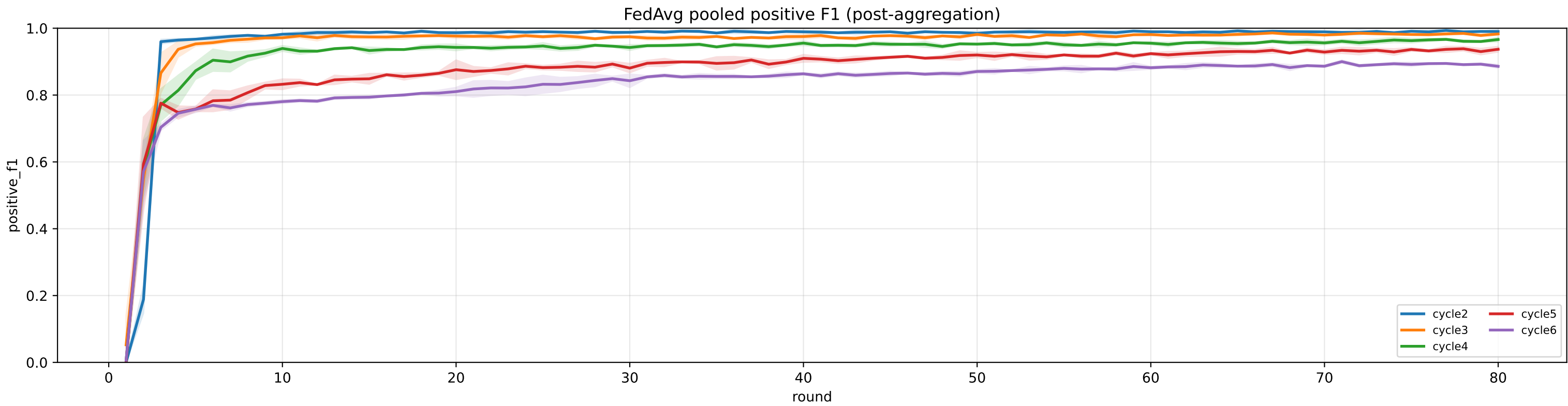
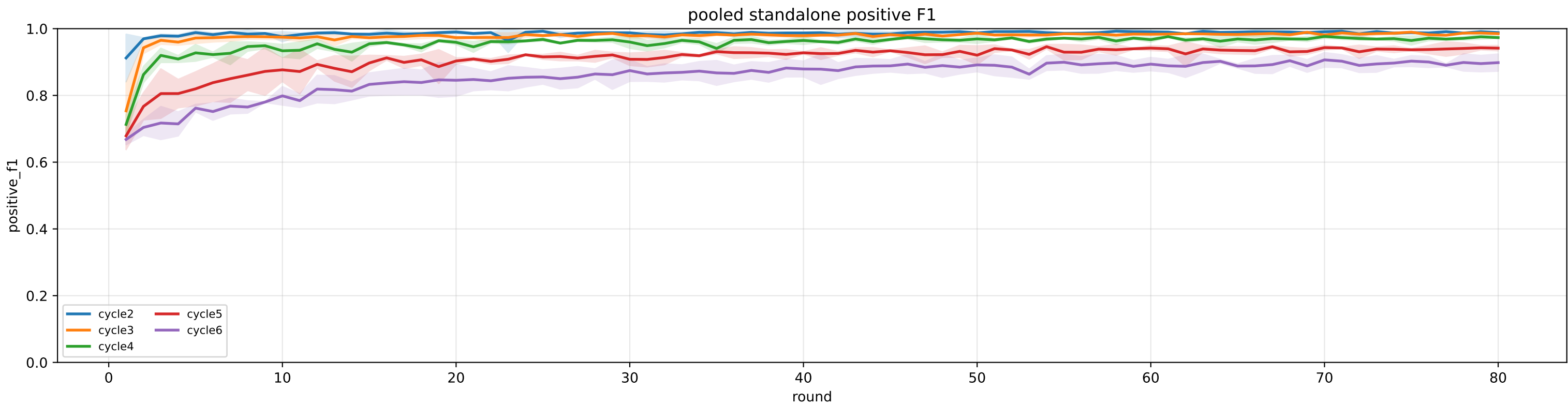
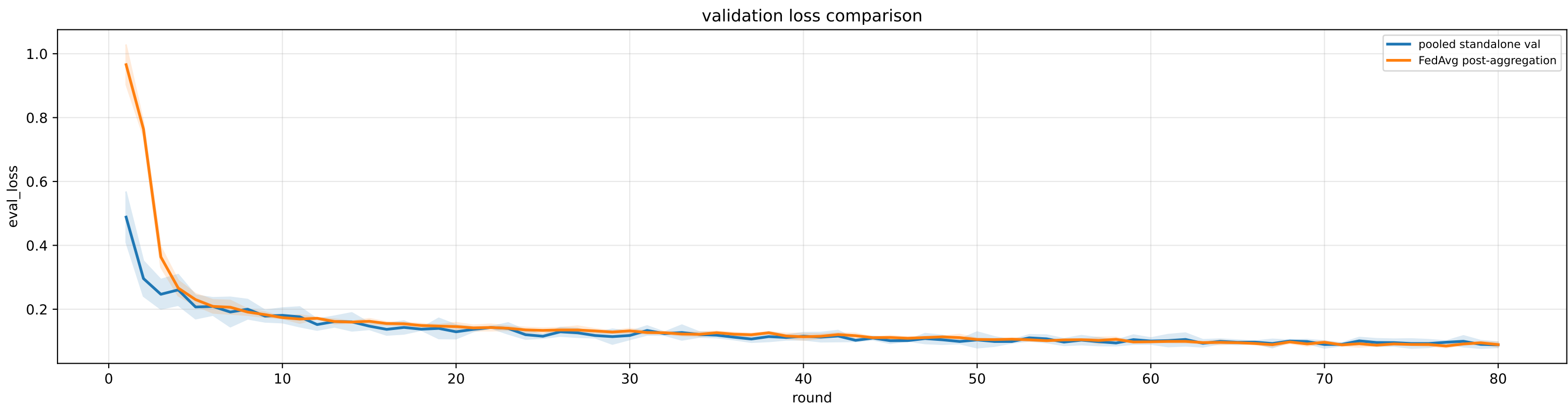
Algorithm	Subset Acc	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	89.1% ± 0.6%	95.0% ± 0.2%	95.5% ± 0.1%	-	-	-	-	-
FedAvg	89.2% ± 0.5%	95.0% ± 0.4%	95.5% ± 0.4%	-	-	-	-	-



graph_id=145 | cycle2=21.4% | cycle3=24.0% | cycle4=26.5% | cycle5=29.0% | cycle6=30.4%

Best test summary (FedAvg vs Local)

Algorithm	Subset Acc	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	90.2% ± 1.3%	95.3% ± 0.6%	95.7% ± 0.5%	-	-	-	-	-
FedAvg	89.4% ± 0.2%	95.0% ± 0.1%	95.4% ± 0.0%	-	-	-	-	-

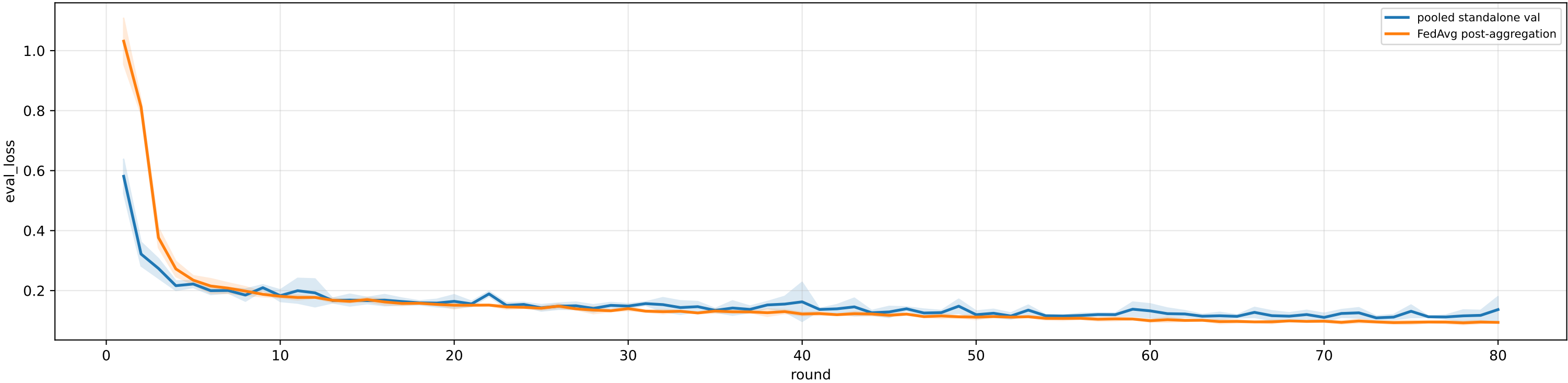


graph_id=166 | cycle2=21.0% | cycle3=24.1% | cycle4=27.1% | cycle5=29.8% | cycle6=28.9%

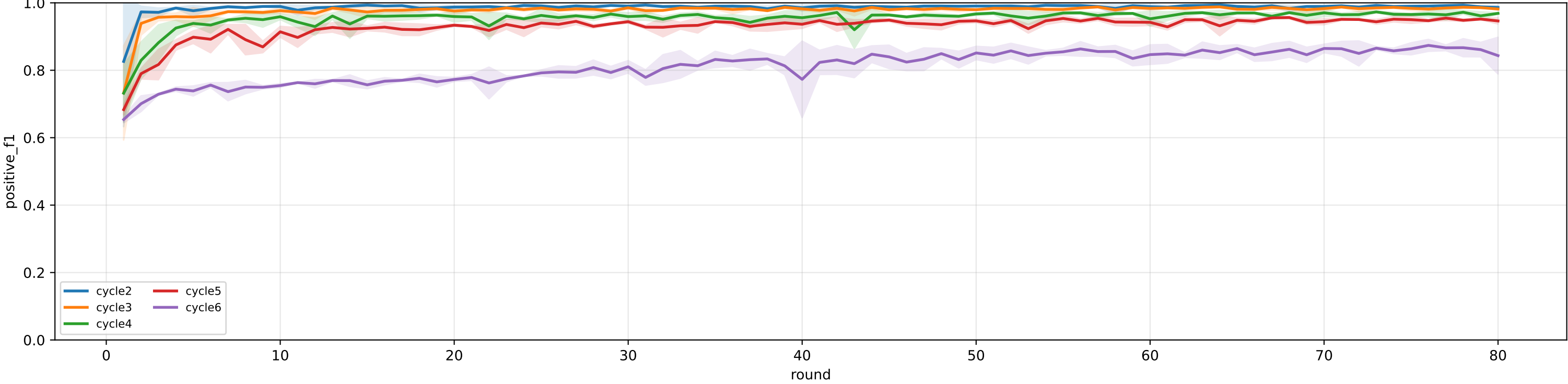
Best test summary (FedAvg vs Local)

Algorithm	Subset Acc	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	87.5% $\pm$ 1.0%	94.6% $\pm$ 0.4%	94.9% $\pm$ 0.4%	-	-	-	-	-
FedAvg	87.6% $\pm$ 0.8%	94.5% $\pm$ 0.3%	94.9% $\pm$ 0.3%	-	-	-	-	-

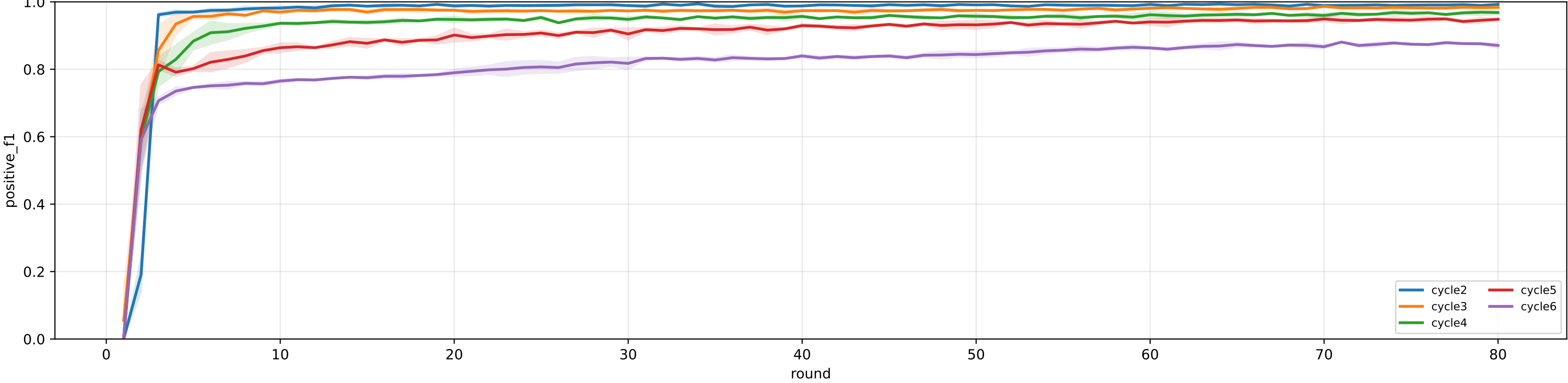
validation loss comparison



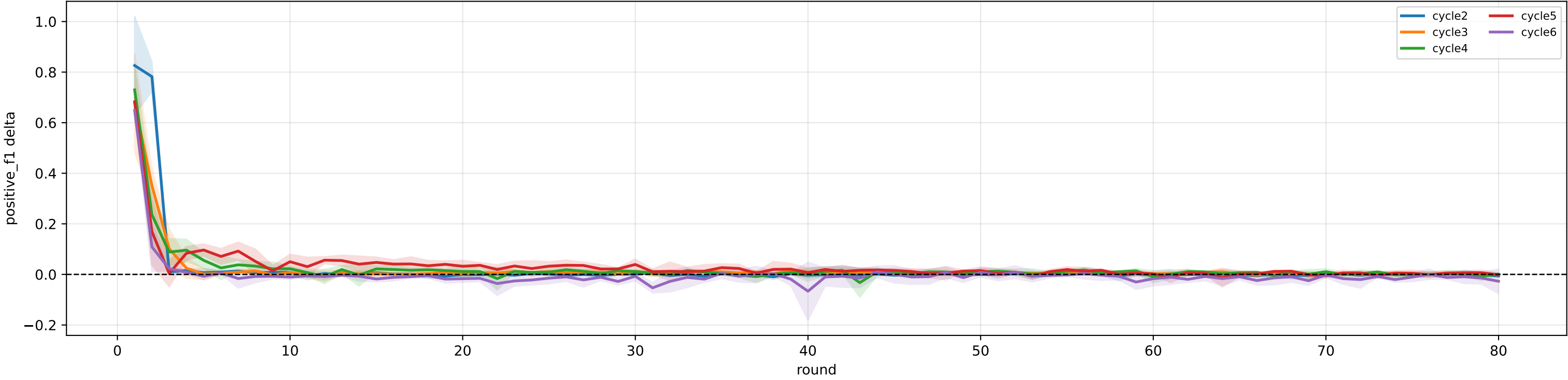
pooled standalone positive F1



FedAvg pooled positive F1 (post-aggregation)



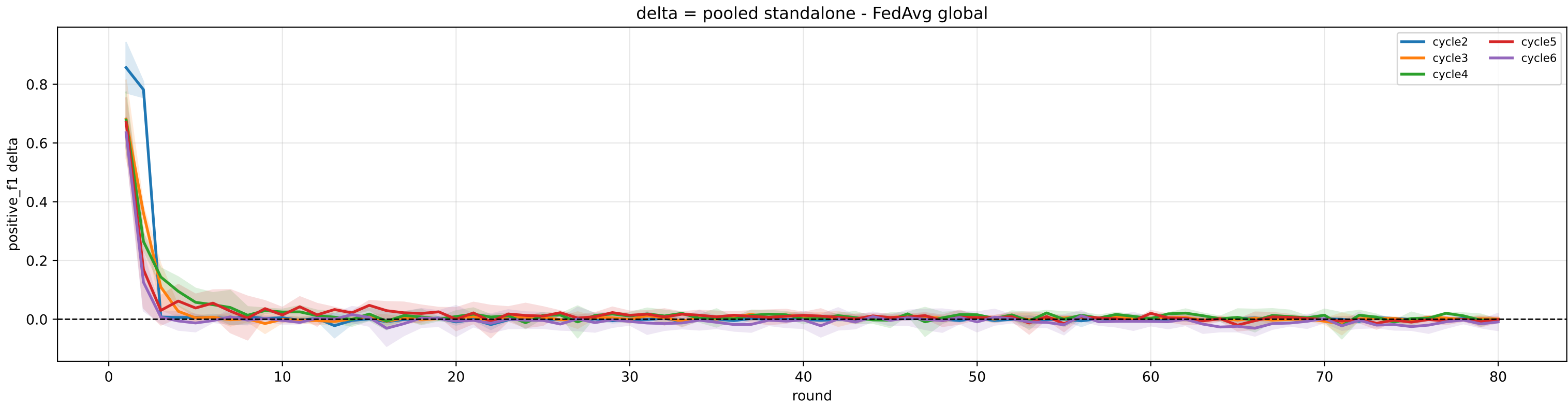
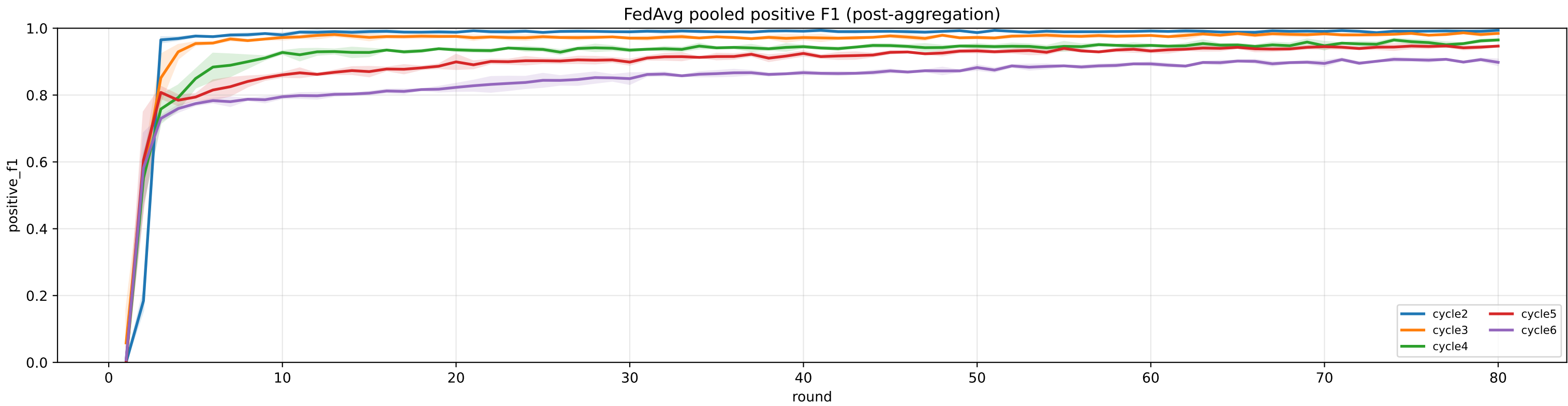
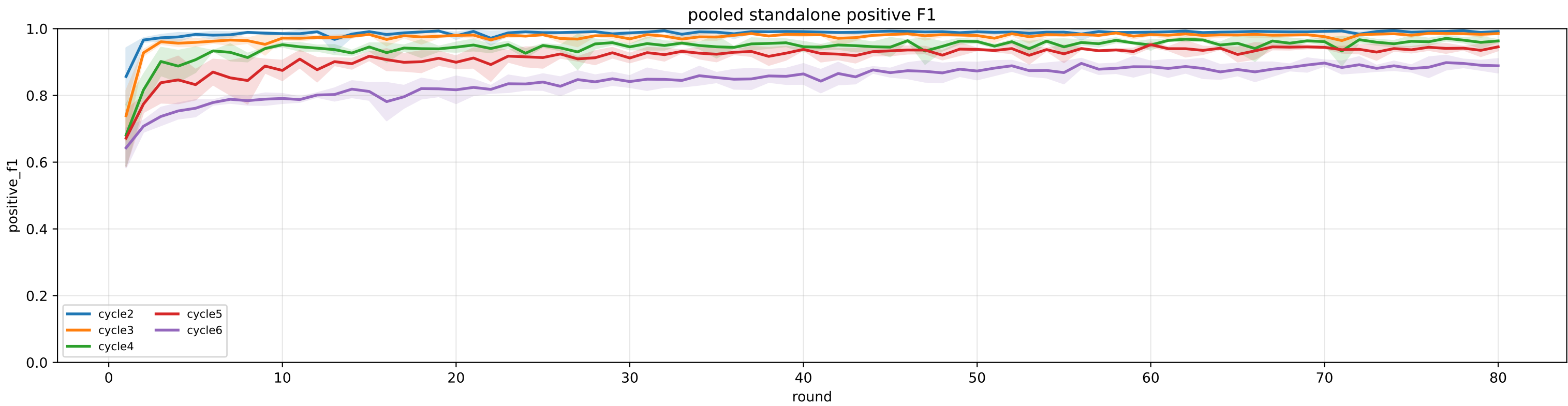
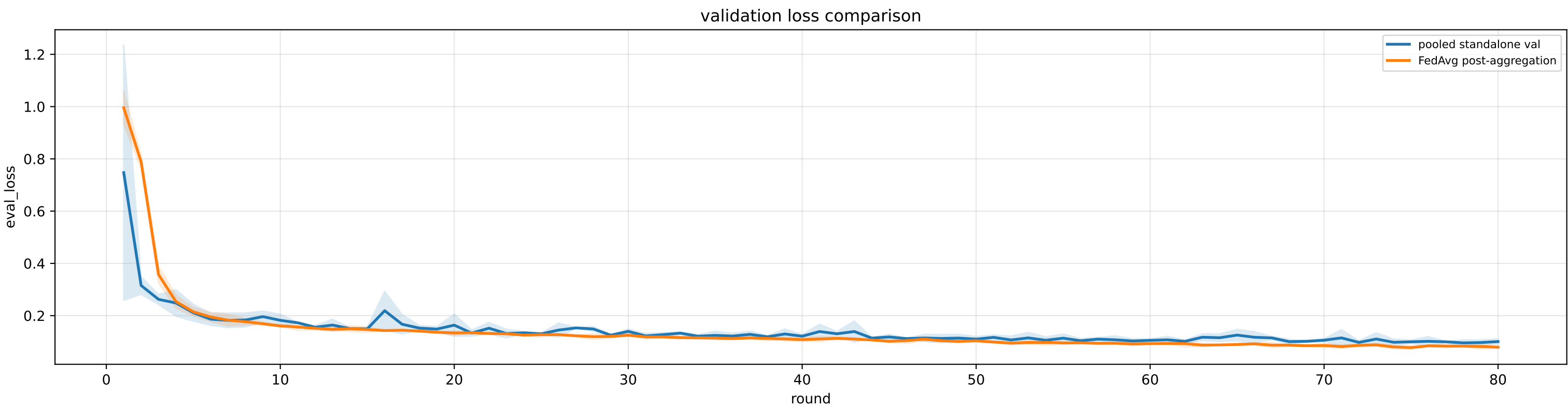
delta = pooled standalone - FedAvg global



graph_id=141 | cycle2=20.1% | cycle3=23.3% | cycle4=25.5% | cycle5=28.7% | cycle6=27.9%

Best test summary (FedAvg vs Local)

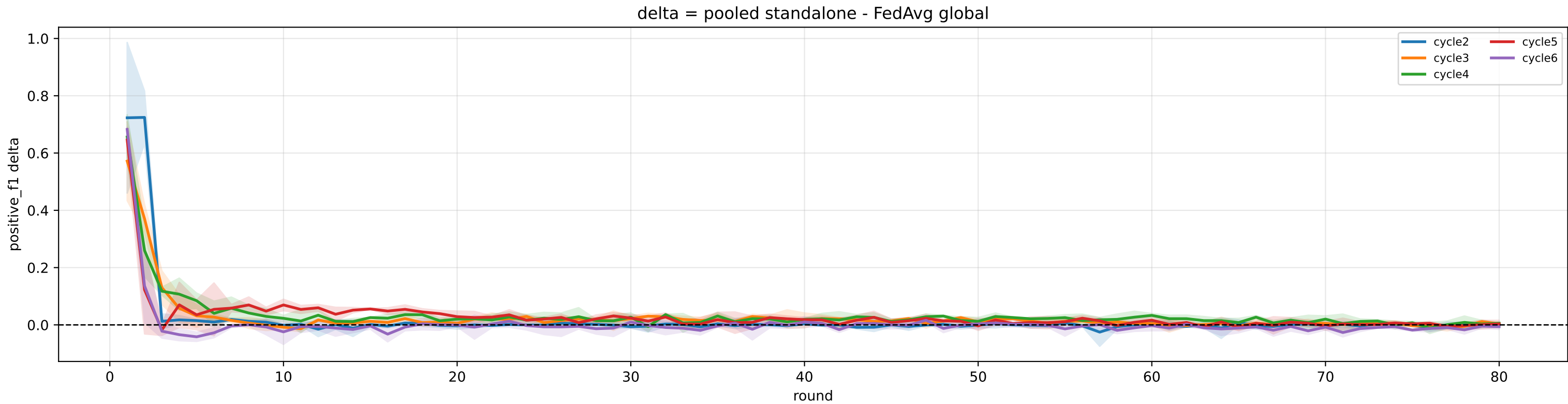
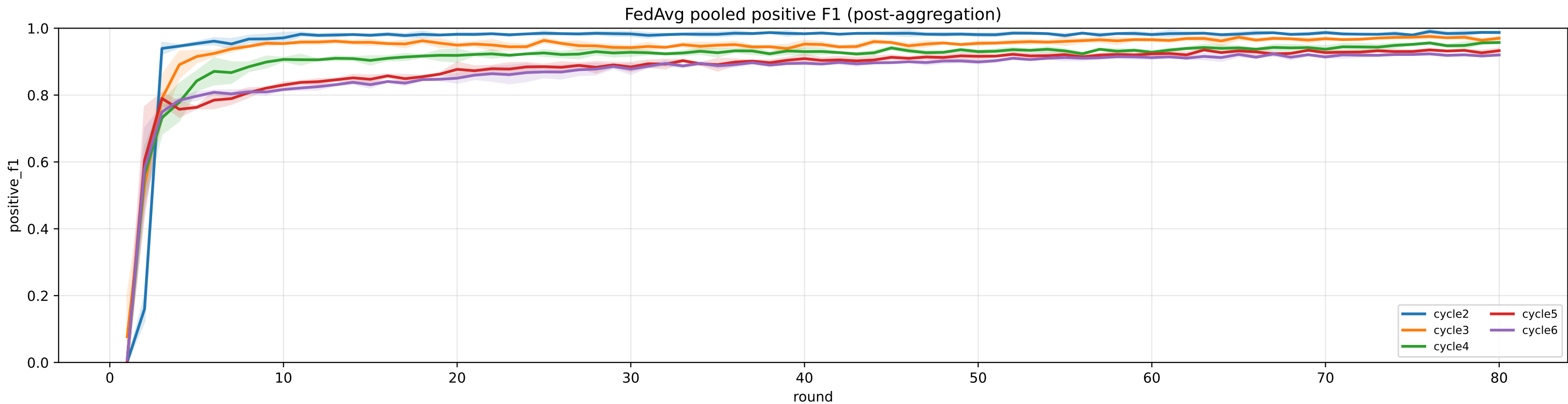
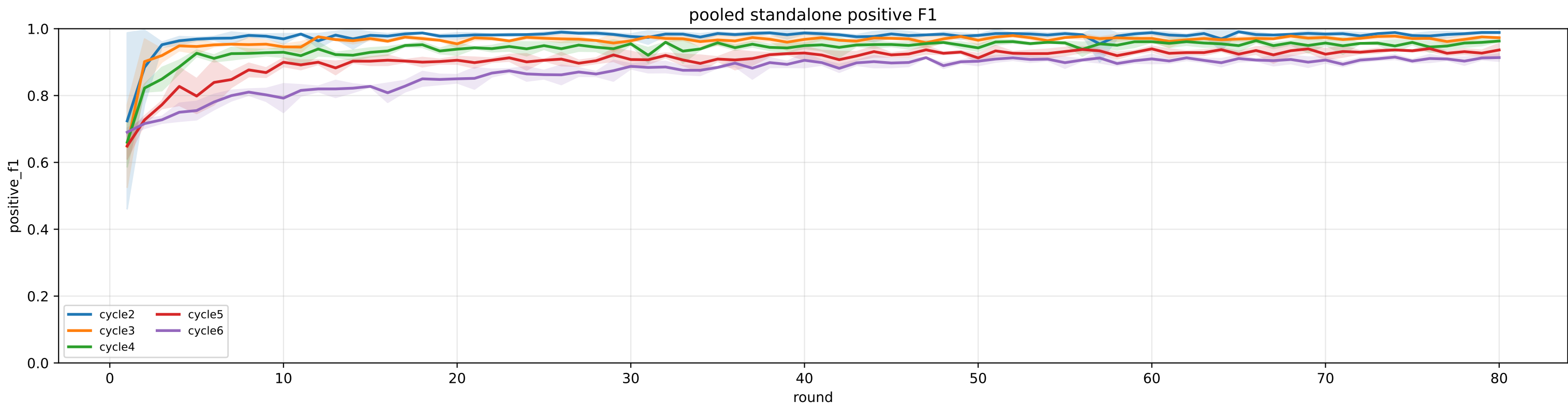
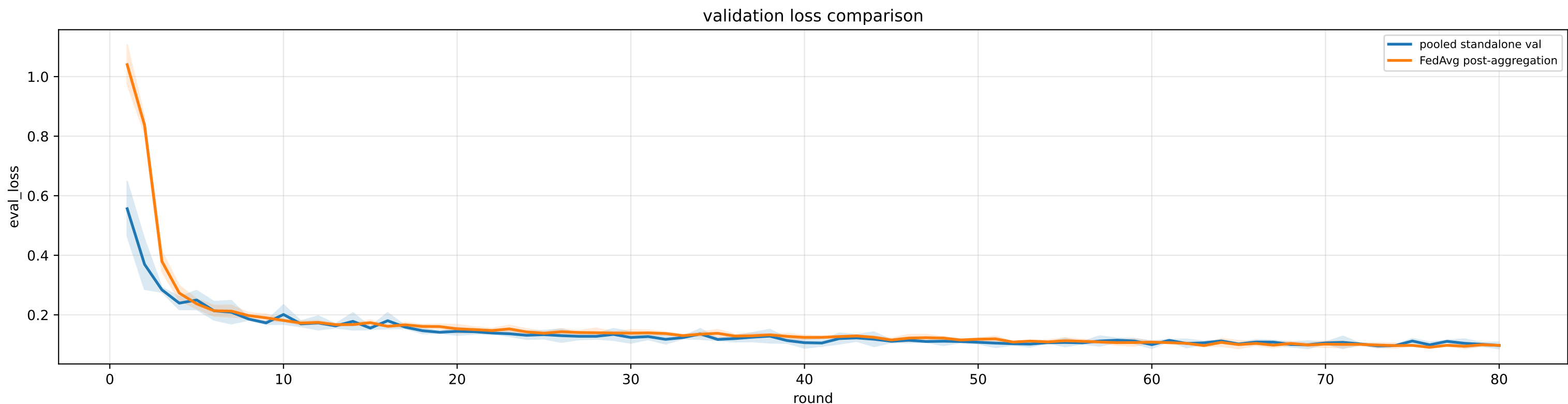
Algorithm	Subset Acc	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	88.7% ± 0.9%	94.8% ± 0.2%	95.3% ± 0.2%	-	-	-	-	-
FedAvg	89.1% ± 0.5%	95.0% ± 0.2%	95.4% ± 0.1%	-	-	-	-	-



graph_id=96 | cycle2=14.0% | cycle3=18.2% | cycle4=22.1% | cycle5=26.3% | cycle6=29.6%

Best test summary (FedAvg vs Local)

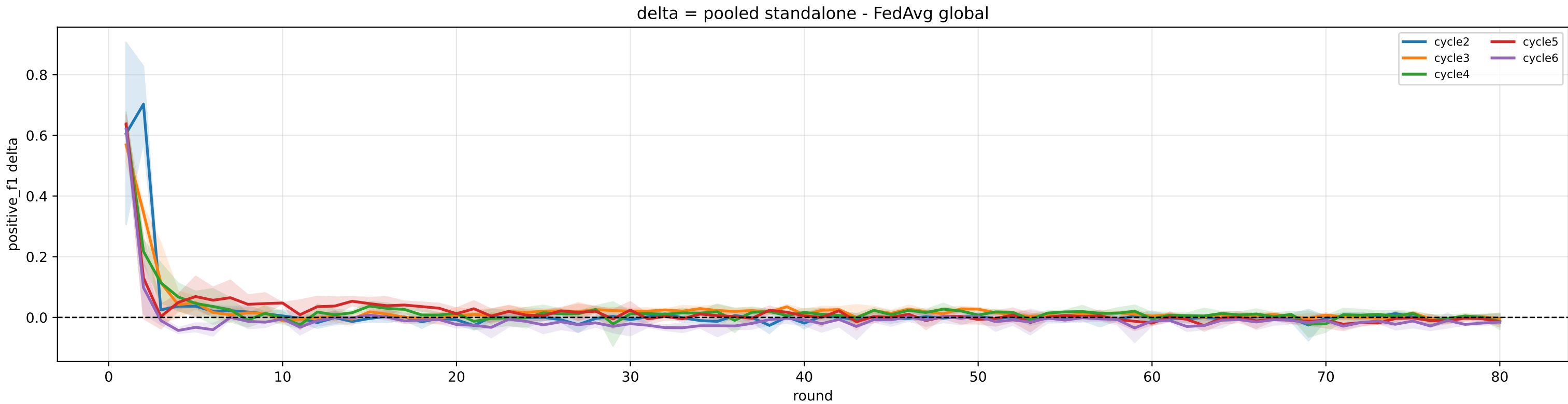
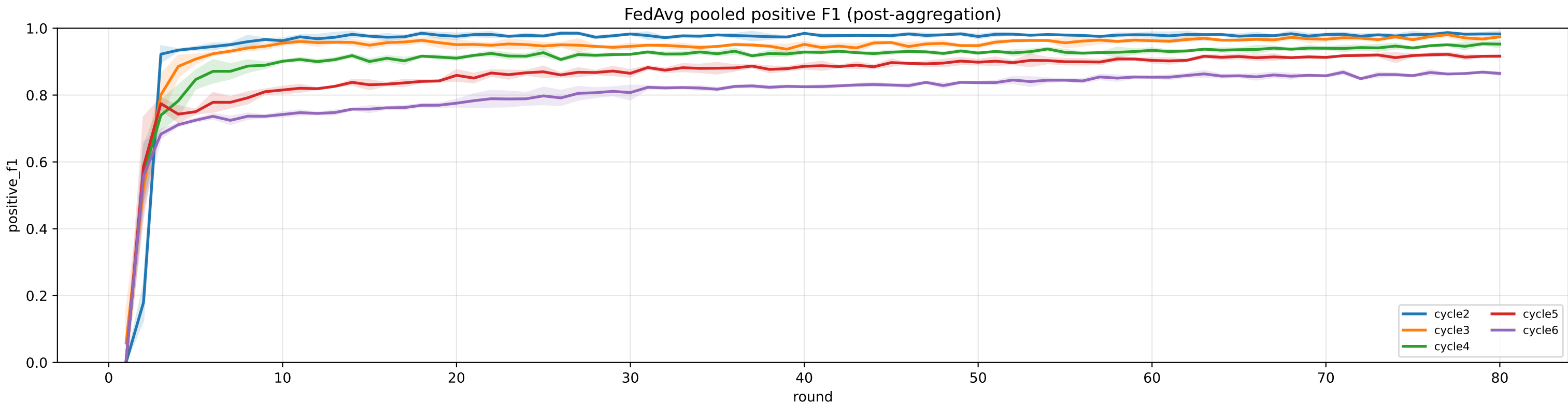
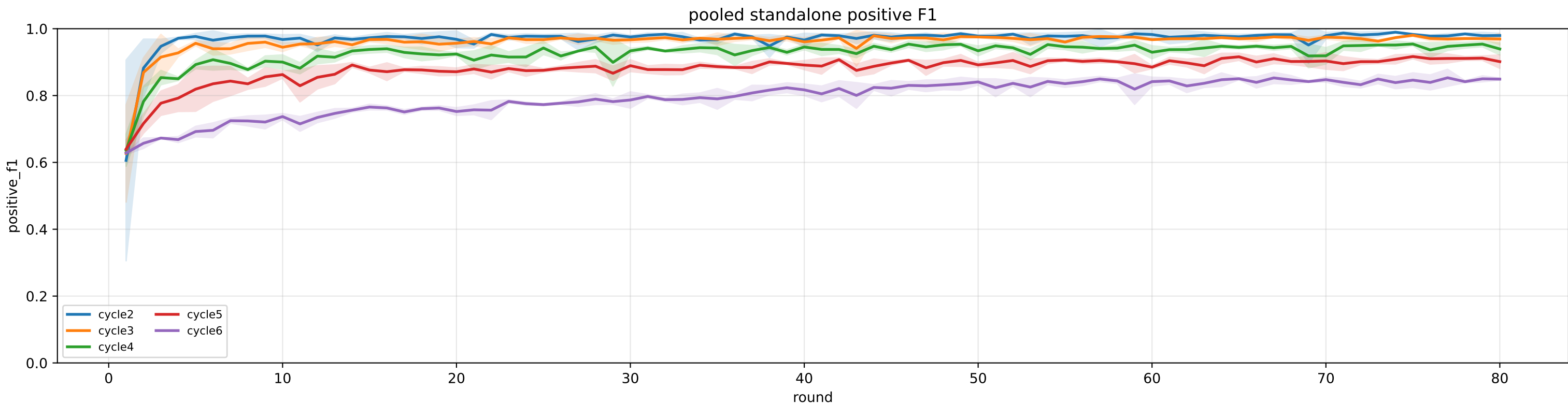
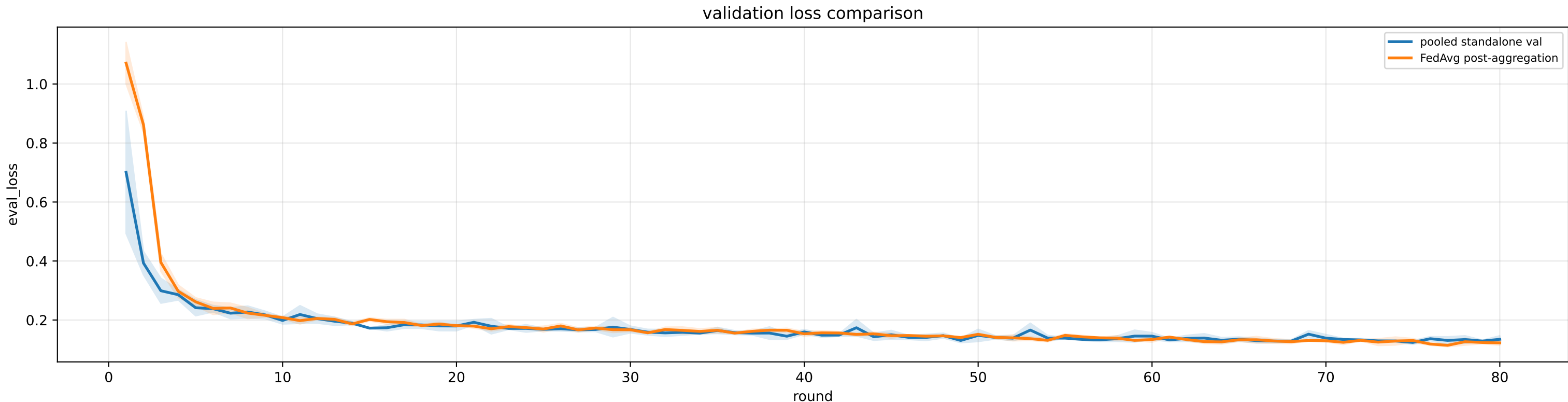
Algorithm	Subset Acc	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	87.1% ± 1.0%	93.3% ± 0.5%	94.4% ± 0.4%	-	-	-	-	-
FedAvg	88.3% ± 0.2%	93.8% ± 0.2%	94.6% ± 0.2%	-	-	-	-	-



graph_id=162 | cycle2=12.9% | cycle3=16.7% | cycle4=22.3% | cycle5=26.0% | cycle6=28.7%

Best test summary (FedAvg vs Local)

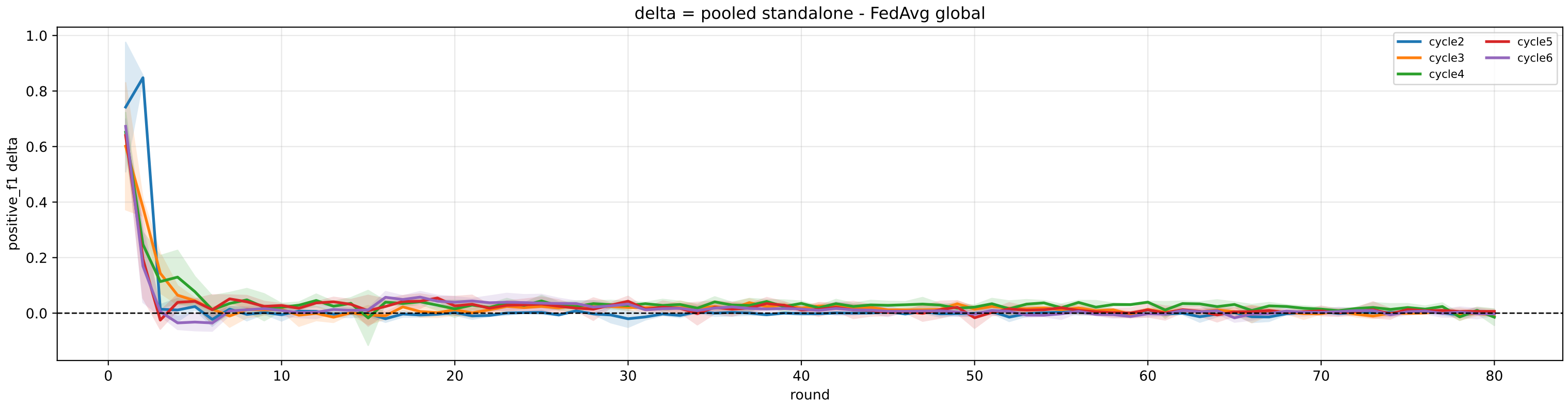
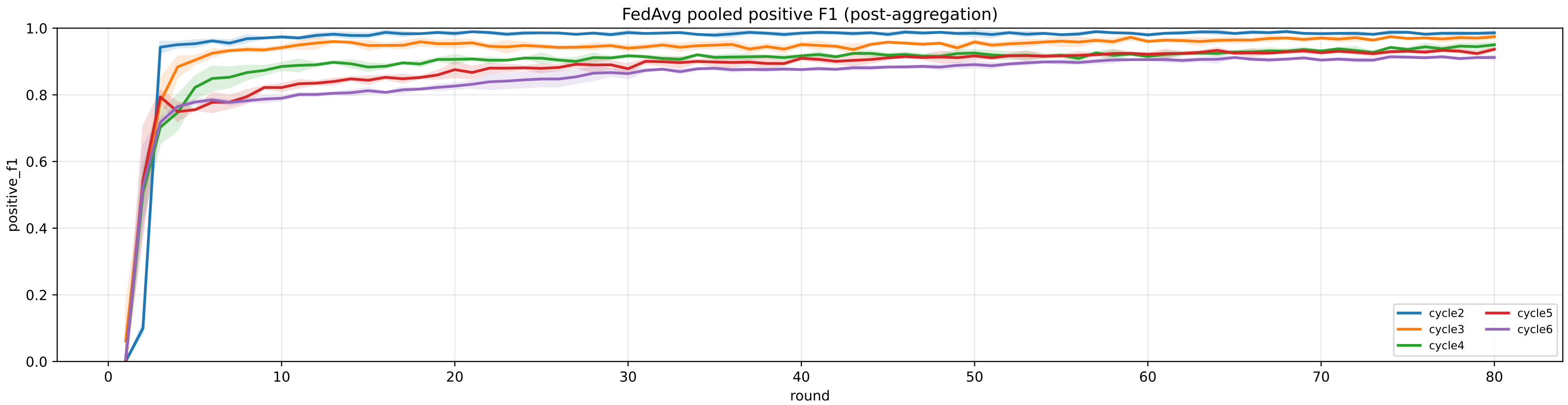
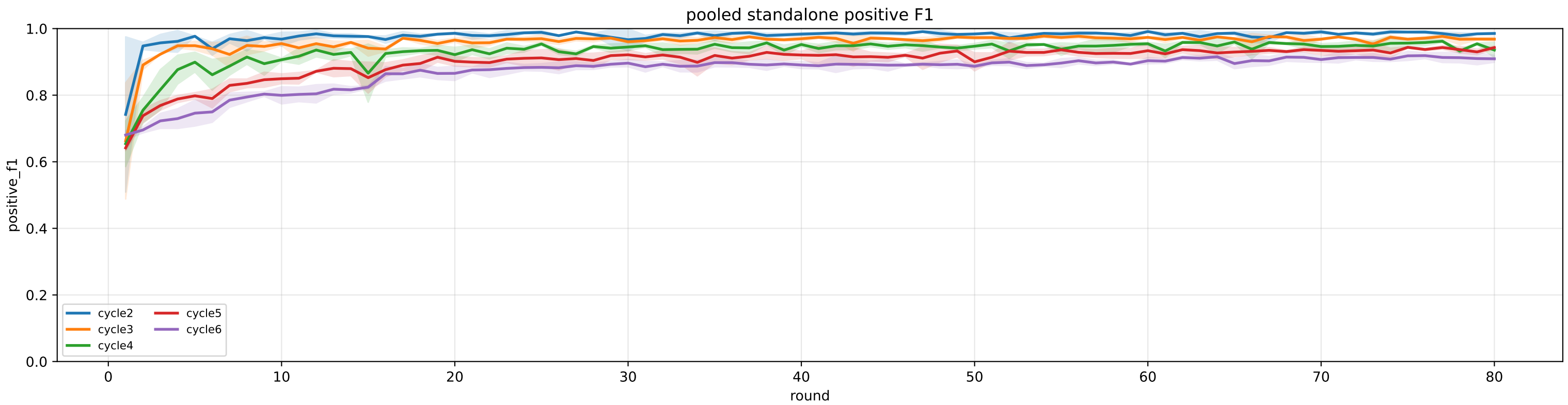
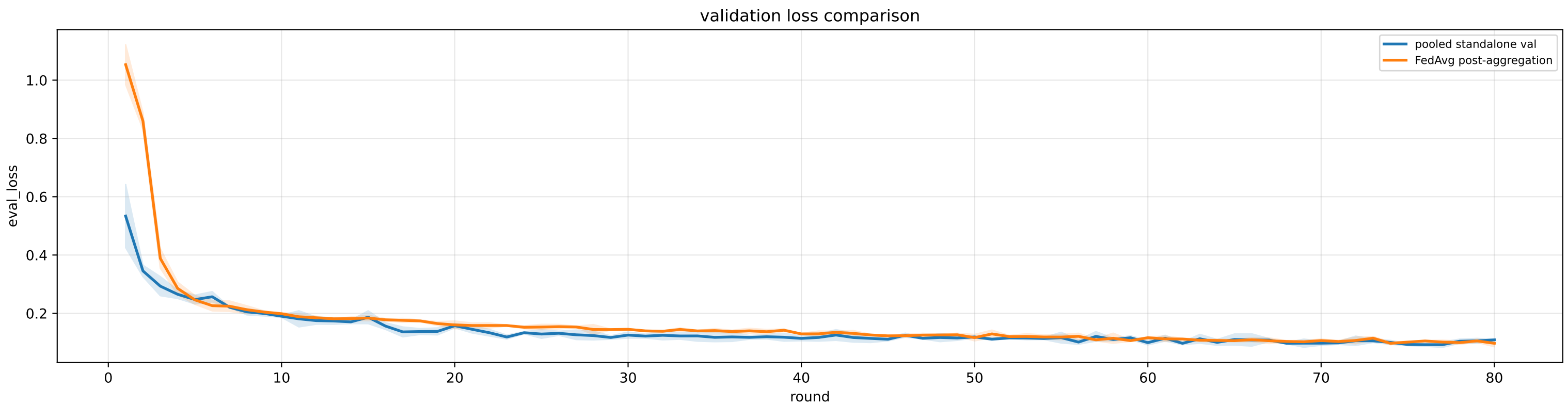
Algorithm	Subset Acc	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	86.2% ± 0.8%	92.8% ± 0.4%	93.8% ± 0.3%	-	-	-	-	-
FedAvg	86.6% ± 0.2%	93.0% ± 0.2%	94.0% ± 0.2%	-	-	-	-	-



graph_id=107 | cycle2=13.7% | cycle3=16.8% | cycle4=21.5% | cycle5=25.0% | cycle6=29.3%

Best test summary (FedAvg vs Local)

Algorithm	Subset Acc	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	88.8% ± 1.7%	94.0% ± 0.8%	95.0% ± 0.7%	-	-	-	-	-
FedAvg	88.2% ± 0.4%	93.6% ± 0.2%	94.6% ± 0.2%	-	-	-	-	-



graph_id=0 | cycle2=13.9% | cycle3=20.5% | cycle4=21.2% | cycle5=27.0% | cycle6=29.7%

Best test summary (FedAvg vs Local)

Algorithm	Subset Acc	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	88.5% ± 0.8%	94.1% ± 0.3%	94.9% ± 0.1%	-	-	-	-	-
FedAvg	88.9% ± 0.4%	94.3% ± 0.2%	95.1% ± 0.2%	-	-	-	-	-

